

Rule-Rewriting Cellular Automata and the Edges of Chaos

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Abstract

We study a two-layer cellular automaton in which each cell carries both a phenotype state and an eight-bit rule. A permutation σ of the eight displayed truth-table positions defines an operator P_σ on those rules by $\text{bit}[\sigma(p)] := \neg \text{bit}[p]$. The resulting family contains $8! = 40,320$ permutation-propagators and is exhaustively classifiable.

The exact, machine-checked classification is that P_σ has a fixed rule exactly when every cycle of σ is even; an all-even permutation with k cycles has 2^k fixed rules; and every fixed rule is balanced, with Langton activity $\lambda = 1/2$. In a census of the spatially coupled dynamics, every odd-cycle operator is genotype-alive under the stated finite-run protocol, whereas every all-even cycle type contains both sustaining and collapsing operators.

Within the wrap-rotation subfamily, coprime rotations sustain high diversity while the other nontrivial rotations collapse in the runs tested. A finite-size scan finds a broad, drifting crossover rather than evidence of a sharpening critical point. Filamentary boundaries show elevated genotype rewriting and, against shuffled and mask-matched nulls, carry directed transport and local combining that are both elevated in an intermediate band of a sampled synthetic order–chaos family; under strong noise the transport ceases to be specific to the filaments. Temporal braids and a phenotype-reading construction provide two reproducible ways to sustain structure beyond the bare operator.

Keywords: cellular automata, rule evolution, Langton’s lambda, self-modifying

systems, edge of chaos, artificial life, permutation operators, exhaustive classification,
domain walls

Two Edges

In 2015, working on a different problem, we built a cellular automaton whose cells carried rules rather than states (Corneli & Maclean, 2015). A programming error connected with one example generated spacetime diagrams reminiscent of the edge of chaos behaviour observed in classical CAs. The question of how that example recovered the distinctive visual behaviour remained open.

Recent computational experiments have shown that that accidentally discovered operator is related to a family of operators that can be given a concrete description, and whose emergent behaviour has interesting properties. A permutation σ of the eight truth-table positions of an elementary rule defines an operator P_σ that rewrites a rule by negating the bits σ names; there are $8! = 40,320$ such operators, and they admit exact classification rather than sampling. Asking this question at the level of this class of operators rather than all possible (8^8) operators gives a more manageable problem.

Asking it turns up two distinct “edges of chaos”. The first is the classical one: a location in *parameter space*, aligned with an order parameter which can be tuned so that the dynamics sit between order and disorder (Langton, 1990). Our classification lands on this critical point from an unexpected direction—every fixed rule of every operator in the family is balanced, with Langton activity $\lambda = 1/2$, exactly the value the classical picture associates with the transition. However, searching the Meta Cellular Automaton (MetaCA) for a critical transition does not produce one (Empirical finding 8). The algebra places the fixed points precisely at the classical critical value; the dynamics find no critical point of this sort at all.

The second “edge of chaos” is not a location in parameter space at all. Because the genotype layer varies from cell to cell, one lattice can hold both ordered and disordered regions *simultaneously*. Tinting the genotype by activity resolves the field into coexisting low- and high-activity domains, and the boundaries between those domains trace fractal-like filamentary walls through spacetime. We observe that these walls are where the

genotype is rewritten fastest, and—against a size-matched null, a geometry-preserving shift null, and a cross-seed null—they carry directed transport that a filament-shaped region placed anywhere else does not. This edge is topographical: a feature of the picture rather than a critical point for the parameter.

Indeed, the spatial walls carry the most transport when the global parameter sits in an intermediate band. They lose their specificity when noise dissolves the domains that define them. A parameter-space region in which physical-space boundaries become computationally active is a two-level claim, and neither level states it alone.

This paper presents a classification of the operators in question, a census of the spatially coupled dynamics across all cycle types, and the interface analysis that produced the second edge. The exact results hold for the entire family. The empirical claims are based on an examination of one synthetic order–chaos family, and one estimator at history length one.

The Object

A cellular automaton normally evolves a field of *states* under one fixed rule. Here each cell also carries a *rule*, and local writes evolve that rule field alongside the phenotype. We display an elementary cellular automaton (ECA) rule in Wolfram order,

$$111, 110, 101, 100, 011, 010, 001, 000,$$

and index these displayed positions from left to right by $p = 0, \dots, 7$. Thus position 0 is the 111 output and most significant bit, while position 7 is the 000 output and least significant bit. Throughout, the indices of σ , w , g , and $\text{bit}[p]$ are these *displayed positions*, not the binary values of the neighbourhoods. If $n \in \{0, \dots, 7\}$ is a numerical neighbourhood value, its rule output is therefore $g(7 - n)$.

We study the operator P_σ , parameterised by a permutation $\sigma \in S_8$ of the displayed positions. Writing the displayed byte as a map $g: \{0, \dots, 7\} \rightarrow \{0, 1\}$, the operator sends g

to $P_\sigma g$ defined by

$$(P_\sigma g)(q) = \neg g(\sigma^{-1}(q)), \quad \text{equivalently} \quad \text{bit}[\sigma(p)] := \neg \text{bit}[p]. \quad (1)$$

Each output bit is one input bit, at the position σ moves it to, negated; P_σ is thus a permutation of the eight coordinates composed with a global complement. The truth table is thereby itself an eight-cell automaton whose wiring is σ and whose update is Boolean negation — an automaton inside the automaton. Equation (1) defines a family of $8! = 40,320$ operators that is finite, exhaustively enumerable, and, as we show below, exactly classified.

The legacy engine stored its table in the order 000, 001, 010, 100, 011, 101, 110, 111. Whenever a legacy writing is imported, the reproduction code conjugates it into the displayed Wolfram order above before we quote its permutation or apply names such as offset +2.

Definition 1 (Writings, elementary writes, and propagators). *A writing is any function $w: \{0, \dots, 7\} \rightarrow \{0, \dots, 7\}$ on the truth-table positions. Its elementary update at source position p is*

$$(U_{w,p}g)(q) = \begin{cases} \neg g(p), & q = w(p), \\ g(q), & q \neq w(p). \end{cases}$$

Thus one update complements source bit p and writes it to target $w(p)$. If w is a bijection—a permutation $\sigma \in S_8$ —all eight writes can be performed simultaneously without collisions, giving the permutation-propagator P_σ in Equation (1). Its fixed bytes are exactly the bytes fixed by every $U_{\sigma,p}$. For a non-injective writing, individual updates $U_{w,p}$ remain well defined, but a simultaneous P_w requires a collision-resolution convention; such a writing also has gaps, or positions targeted by no source.

Observation 1: Permutations form a small bijective core

Permutations are only the bijective core of a far larger class: there are $8! = 40,320$ permutations but $8^8 = 16,777,216$ writings, so the permutations are 0.24% of the whole. The cycle-structure algebra of *The Classification* is available only on this core. The historical operator $[00123456]$ (Figure 3) is itself a non-bijective writing—bits 0 and 1 both write to position 0, and position 7 is never written—and its behaviour ranges widely between its two standard-order forms.

Box 1 works through one member of the family bit by bit.

Worked example 1: A simultaneous propagator by hand

Write an ECA rule in Wolfram’s standard neighbourhood order:

position	0	1	2	3	4	5	6	7
neighbourhood	111	110	101	100	011	010	001	000
Rule 110	0	1	1	0	1	1	1	0

Thus Rule 110 is the byte 01101110. Let $\sigma(p) = p+2 \pmod{8}$ on the displayed position indices: each source bit is complemented and written two positions further to the right, wrapping at the end. Carrying out the eight writes gives

$$P_{+2}(01101110) = 01100100,$$

which is Rule 100. This changes what the cell does to its black-and-white phenotype. For example, neighbourhood 011 maps to 1 under Rule 110 but to 0 under Rule 100. The propagator has not directly flipped a phenotype cell; it has changed the truth table that the cell will use.

The simultaneous operator acts on one eight-bit rule; the spatial experiments place its elementary writes inside a two-layer automaton.

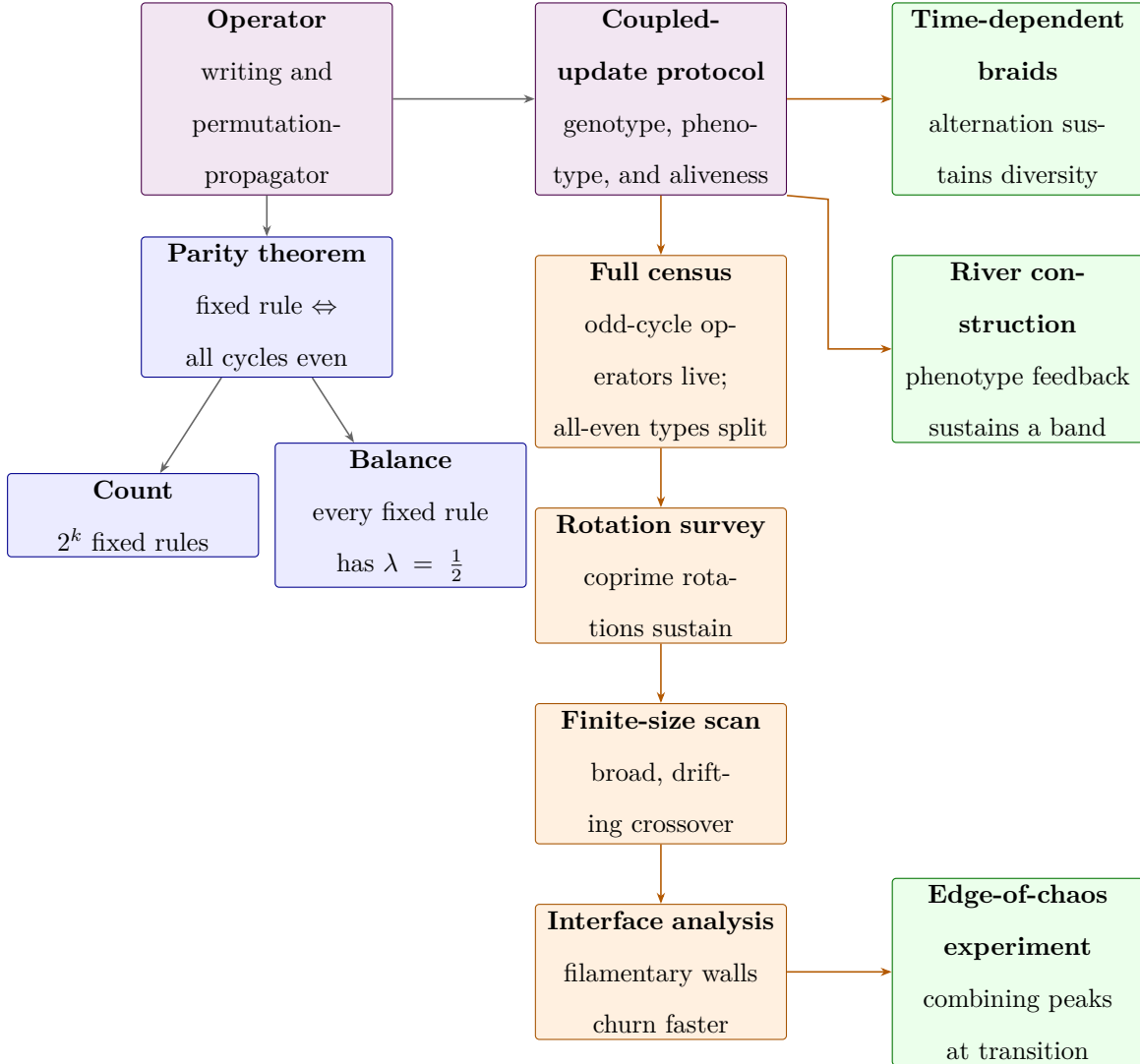


Figure 1

Argument map. The exact branch (blue) derives fixed-point existence, number, and balance from the operator definition. The empirical branch (amber) uses the coupled-update protocol to move from the full census to the rotation survey, finite-size scan, and interface analysis. The forward branch (green) tests two ways of sustaining structure and a sampled edge-of-chaos computation experiment. Arrows mark logical or methodological dependence; figures visualise the corresponding measurements but are not themselves premises.

Definition 2 (Genotype and phenotype). *Each cell carries an eight-bit rule. The field of rules—the genotype—governs a field of black-and-white states—the phenotype—and neighbouring rules also interact. One complete local update (Box 2) evolves both layers; this is the computation behind the space–time diagrams in Figure 2.*

Worked example 2: One site, one complete MetaCA step

At time t , site x holds an eight-bit rule $g_x(t)$ and a phenotype bit $X_x(t)$. One ordinary coupled step has three parts.

1. **Express the current rule.** The phenotype neighbourhood is evaluated by the rule at its centre. With numerical neighbourhood value $n_x(t) = 4X_{x-1}(t) + 2X_x(t) + X_{x+1}(t)$, its displayed position is $7 - n_x(t)$:

$$X_x(t+1) = g_x(t)(7 - n_x(t)).$$

2. **Blend neighbouring rules.** For each truth-table position j , inspect the three genotype bits $(g_{x-1}(j), g_x(j), g_{x+1}(j))$. If the two outer bits agree, copy their value. Thus $(0, 1, 0)$ blends to 0. If they disagree, the centre rule decides by treating the triple as one of its ECA neighbourhoods, converting its numerical value n to displayed position $7 - n$ as above. The eight coordinate-wise decisions form an intermediate rule

$$h_x = B(g_{x-1}(t), g_x(t), g_{x+1}(t)).$$

3. **Write one propagated bit.** Draw a source position $K_x(t) \in \{0, \dots, 7\}$ uniformly and apply its elementary write to the intermediate byte:

$$g_x(t+1) = U_{\sigma, K_x(t)}(h_x).$$

The ordinary spatial runs make one such elementary write per cell and generation. The historical Figure 8 variant made two. By contrast, P_σ denotes the simultaneous eight-write map classified in *The Classification*; its fixed bytes are precisely the bytes left unchanged by every possible elementary write.

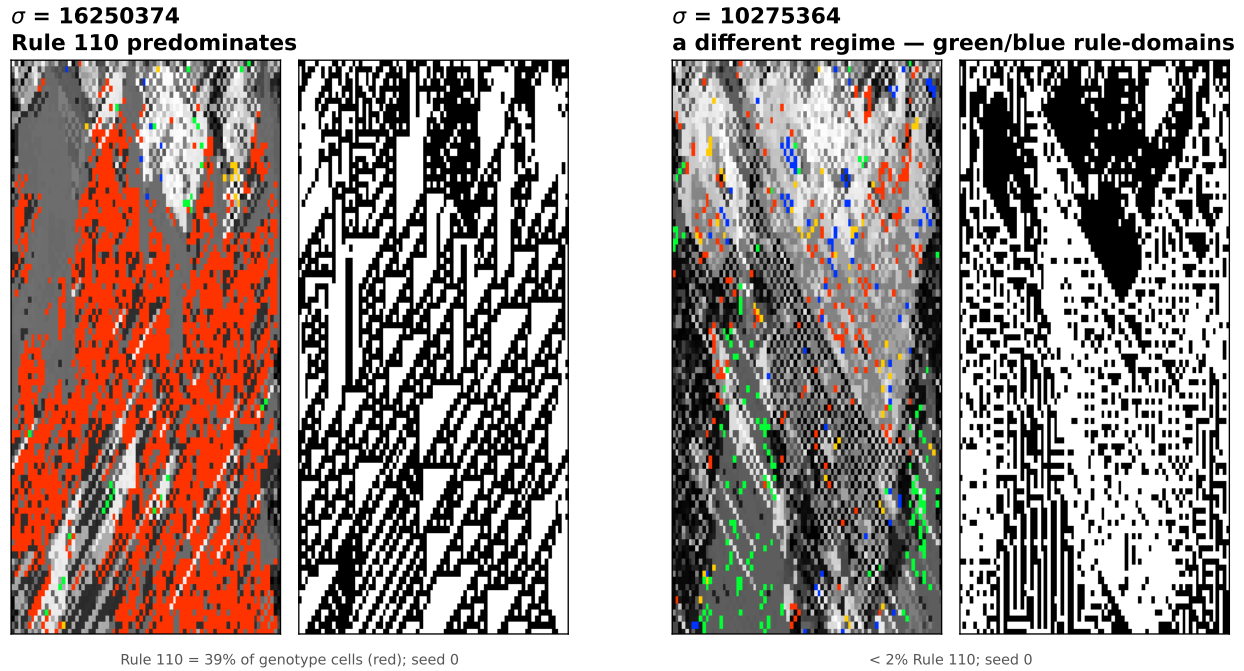
The substrate—a cellular automaton whose cells hold rules rather than states—is not itself new (Corneli & Maclean, 2015); what we study is the *operator* of Equation (1), understood as an enumerated, exactly classified family. That 2015 paper found a single member of this family, encountered through a programming error and not recognised as one of many; we now give the family an essentially complete characterisation. The canonical symmetry group of ECA rule space has order four (reflection $\times$ complement, collapsing 256 rules to 88 classes) (Schaller & Svozil, 2025); it is not S_8 , and the negation-carrying, σ -parameterised family of Equation (1) does not appear in it. Rule-space permutation as a parameterised, enumerated family with a derived fixed-point structure does not appear in the literature we have surveyed; the nearest neighbours augment cellular automata with state-dependent feedback (Pavlic et al., 2014) rather than permuting the rule itself. Two established genres are adjacent but distinct: per-cell rules (non-uniform or ν -CA (Dennunzio et al., 2011)) and rules that change during evolution (rule-changing or programmable CA (Mori et al., 1998)). Non-uniform cellular automata are not, by themselves, the source of novelty here: when each site draws its rule from a finite set, the resulting system can be represented as a uniform cellular automaton on a product alphabet (Dennunzio et al., 2011). Our contribution instead concerns the permutation-propagator family and its fixed-point structure.

The Classification

Fixed points exist iff every cycle is even

Theorem 1. *P_σ has a fixed point (a fixed rule) if and only if every cycle of σ has even length. Since a one-cycle has odd length, this already requires σ to be fixed-point-free.*

A fixed rule is a g invariant under (1), i.e. a σ -equivariant Boolean colouring with $g(\sigma k) = \neg g(k)$. Follow a cycle of length L : equivariance forces $g(k) = (\neg)^L g(k)$, and $(\neg)^L$ is the identity exactly when L is even; conversely, colour each orbit by the parity of its distance from a chosen representative. A one-cycle of σ ($L = 1$) is odd and so admits no

**Figure 2**

Two operators from the family, each shown over time as genotype (256-colour, left) and phenotype (black and white, right). Under $\sigma = 16250374$ the genotype field comes to be dominated by ECA Rule 110—the red cells, 39% of the genotype at the representative seed (35% over six seeds)—and the phenotype accordingly displays Rule 110’s characteristic gliders; under $\sigma = 10275364$ a different regime of rule-domains forms (green and blue, under 2% Rule 110). The operator does not merely stir rule space: it drives it toward particular, computationally distinguished rules. Genotype colours are the legacy engine’s own.

such colouring, which is why even cycles are required throughout.

Box 3 shows the theorem on the offset-+2 operator. The same small calculation also makes the later 2^k count and forced $\lambda = 1/2$ balance visible before we state them generally.

Worked example 3: Even and odd cycles

For offset +2 on eight truth-table positions,

$$\sigma = (0\ 2\ 4\ 6)(1\ 3\ 5\ 7).$$

A fixed rule must change colour at every arrow because $g(\sigma k) = \neg g(k)$. Around either four-cycle it can therefore read 0101 or 1010: after four negations it returns consistently to its starting bit. The two four-cycles are coloured independently, so two binary choices fix all eight truth-table entries $\text{bit}[0], \dots, \text{bit}[7]$. Read back as an ECA rule number—those eight bits in Wolfram neighbourhood order (the table of *The Object*) taken as an integer in 0–255—the four colourings are the fixed rules

$$2^2 = 4 : \quad 51, 102, 153, 204,$$

each with four zero-bits and four one-bits. An odd cycle admits no such colouring: after an odd number of negations the bit returns to its starting position demanding its own opposite, so no fixed rule exists. Admitting a fixed rule and settling onto one are distinct. An operator carrying an odd cycle has no fixed rule for its field to collapse onto, and its field correspondingly stays diverse (the any-odd reduced rotate+2, Figure 5, right); whether a fixed-point-bearing operator collapses onto one of its fixed rules is a separate, dynamical question (*Fixed Points Are Not Attractors*).

This is the signed-cycle fixed-point theorem of Boolean-network theory (Aracena, 2008) specialised to (1): our operator is a Boolean network whose signed interaction digraph is σ with every arc negative, a negation-cycle of length ℓ has sign $(-1)^\ell$, and an isolated negative cycle admits no fixed point while a positive (even) one admits exactly two.

We contribute a machine-checked proof of the specialised statement in Lean 4 / Mathlib.¹

The closed form

The number of $\sigma \in S_{2n}$ with all cycles even is $((2n - 1)!!)^2$, with exponential generating function $\sum_n ((2n - 1)!!)^2 x^{2n} / (2n)! = 1/\sqrt{1 - x^2}$ (the “set of even cycles” construction). For S_8 this is $105^2 = 11,025$ of 40,320, or 27.34%. The count is classical: it is OEIS Foundation Inc. (2011) and appears as Stanley (1999, Problem 5.34(c)) and in Riordan (1968). We claim it only as the enumeration of our operator family, not as a new identity.

Fixed bytes number 2^k

Corollary 1. *An all-even σ with k cycles has exactly 2^k fixed bytes; any σ with an odd cycle has none.*

Each even cycle admits exactly two alternating colourings, chosen independently across cycles, whence 2^k ; the zero case is Theorem 1. This too is machine-checked.² Summed against these weights the family is consistent: $\sum 2^k$ over all-even σ equals $(2n)! = 8! = 40,320$, the total number of fixed bytes across the family.

Every Fixed Point Is Balanced

Definition 3 (Langton’s activity λ). *λ measures a rule’s activity: the fraction of its neighbourhood entries that map to the non-quiescent state (Langton, 1990)—for an eight-bit ECA table, the Hamming weight over eight, $\lambda = \|r\|_1/8$, from 0 (every neighbourhood dies) through 1/2 to 1. A rule with $\lambda = 1/2$ is balanced.*

¹ `DarkTower/Patterns/Propagator.lean`, theorem `hasAlternatingColouring_iff_cycleType_even`; `lake build clean`, no sorry or axiom.

² `alternatingColouring_card_eq_two_pow_cycleType_card`.

Observation 2: Balance is a truth-table property

Langton’s thesis was that complex behaviour concentrates near a critical λ , shown most clearly for larger tables ($K = 4$, $N = 5$); for the two-symbol, three-neighbour ECAs Langton is explicit that λ only roughly correlates with dynamics, and later work is cautious (Mitchell et al., 1993). Theorem 2 below concerns the *value* $\lambda = 1/2$ itself—balance—which we keep separate from any empirical claim about what it predicts dynamically.

Every fixed point has $\lambda = 1/2$

Theorem 2. *Every fixed point of every operator in the family has exactly four one-bits; equivalently, Langton’s activity parameter $\lambda = 4/8 = 1/2$.*

By Theorem 1 each cycle has even length $2m$; an alternating colouring assigns such a cycle exactly m ones; summing over cycles gives $(\sum L)/2 = 8/2 = 4$. Exhaustive enumeration confirms it: across all 40,320 operators and all their fixed points, every one has Hamming weight four, with zero exceptions. The statement is machine-checked for Fin 8.³

We are deliberate about the standing of this result. Langton’s critical value was obtained empirically, as a statistic over rule space (Langton, 1990), and its status as an order parameter was contested: Mitchell et al. (1993) refuted Packard (1988)’s specific genetic-algorithm clustering claim, not the definition of λ , which survived as one axis among others such as Wuensche’s Z (Wuensche, 1999). There is a folklore intuition for why $1/2$ is special — binomial sampling variance $\lambda(1 - \lambda)$ is maximal there — and it is not ours; and it has been noted that λ alone does not fix a rule’s dynamical class, distinct classes coexisting at a single value (Sakai & Kanno, 2002). What Theorem 2 adds is narrower and firmer than any of these: under the complementation in (1) the value $1/2$ is *forced* as an algebraic matter, because any alternation-under-complement colouring is

³ `alternatingColouring_true_card_eq_four`: the true-cells and false-cells are put in bijection by σ , forcing each to number four.

balanced by parity. We claim balance, exactly; any connection of that balance to dynamical behaviour we treat as interpretation, not as part of the theorem.

The cast: which rules survive, and how many

Under an all-even σ , the fixed points of P_σ —the rules it maps to themselves—number 2^k , where k is the number of cycles of σ (*The Classification*): each even cycle admits two alternating colourings, chosen independently across cycles. In the spatial runs these are the rules onto which the field is observed to settle. Every one has $\lambda = 1/2$ (Theorem 2), so the observed terminal population is a cast of *distinct* rules—distinct truth tables—sharing a *single* activity.

The smallest nontrivial case of the count is a single cycle, $k = 1$: two survivors. The 2015 paper’s Figure 8 already exhibited a two-rule residual: a random field of tens of distinct rules dies off to an alternating population of just two, rules 42 and 170. Fittingly, the programming error that produced the family is what pins its residual at two—the erroneous mutation froze a bit, splitting the field into the cells that can reach 42 and those that can reach 170—so the phenomenon was on the page before the 2^k count explained it.

That die-off curve—the count of distinct rules present over time—is a natural population-level companion to λ . Where λ is a scalar assigned to a single rule, this curve is a property of the whole field’s trajectory; and where λ is uninformative here—every fixed point sits at $1/2$ —the curve still varies, because it measures the population rather than the point. We use it cautiously, as an observable rather than an established order parameter, to tell the rotations whose fields stay diverse from those whose fields collapse (*Fixed Points Are Not Attractors*, Figure 7).

Diversity of the observed population and singularity of its activity thus coexist; the metric measures we tried do not separate the two (*Limits of the Tested Distance and Complexity Measures*).

A dead genotype can carry a live phenotype

The balanced shell is a definite object: exactly $\binom{8}{4} = 70$ rules carry four active outputs, and by Theorem 2 the absorbing set of every operator in the family lies wholly inside it. Collapse is therefore not merely a loss of diversity but a fall onto this particular 70-rule balanced shell at $\lambda = 1/2$.

Balance is necessary for a genotype to freeze and silent about what the frozen rule then does. Run each of the 70 as an ordinary elementary CA and they split: 32 are phenotypically live—sustained change under iteration, the additive rules 90, 105, 150 and the class-IV rule 54 among them—while the rest fall quiescent. The two halves share a single λ ; this is, inside our family and by construction rather than by survey, the coexistence of dynamical classes at one activity value that has long been urged against λ as an order parameter (Sakai & Kanno, 2002). A field can thus die in its genotype while the surviving rule keeps its phenotype in motion.

Figure 4 is such a field, built to specification. To seat a collapse on a chosen live rule R it suffices to pair R 's four active positions with its four inactive ones as an involution: four transpositions, an all-even operator that holds R among its fixed points (Theorem 1). Whether the field actually settles onto that fixed rule is a separate, dynamical question (as *Fixed Points Are Not Attractors* stresses); here it does. Carried out for Rule 54 this yields $\sigma = [2\ 3\ 0\ 1\ 5\ 4\ 7\ 6]$; run from a random field it drives the genotype to near-uniformity—two rules survive—while the phenotype never settles. The field has died onto Rule 105, another live tenant of the same shell, whose additive dynamics keep the black-and-white panel full of structure. Genotypic death and phenotypic death are not the same event.

One caveat marks the open edge of this. An involution fixes not its target alone but the whole shell compatible with its pairing, and which tenant a random field actually reaches is set by the basins, not by the construction: we aimed at 54 and the dynamics preferred 105. Existence is settled—live survivors are constructible—but *steering* a field onto a named rule is not, and is the natural next question.

The Census

The parity dichotomy in the spatial dynamics

Theorems 1–2 concern fixed bytes; the spatial dynamics adds stochastic elementary writes and, in the census, neighbour blending. Figure 5 shows why fixed-point existence is not itself a dynamical verdict: without blending, all-even rotate+2 settles within the run while all-even rotate+1 remains diverse; the odd-cycle examples, which lack fixed bytes, also remain diverse. The full census below (*The full census, tabulated by class*) makes this quantitative over all 40,320 operators (Table 1); the dynamical content is the *death-rate difference* between the classes, not the class sizes.

A prospective test: reduced rotate+2

To guard against reading structure into a completed enumeration, we fixed a prediction before running the case. The reduced rotate+2 operators act as a single even cycle on a subset of positions and leave the remaining positions as one-cycles; by that leftover they are *not* all-even, so by Theorem 1 they possess no fixed point and should fall on the no-fixed-points side of the dichotomy—no rule for the field to settle onto, and correspondingly less extinction. They do (Figure 5, right): they stay diverse rather than dying off. Absence of fixed points forbids settling onto a single rule; it does not forbid short-period behaviour, and we claim only the observed persistence of diversity, not literal aperiodicity.

The full census, tabulated by class

Definition 4 (Census protocol and aliveness). *For each of the 40,320 permutations, we ran seeds 0–2 on a width-60 line for 100 generations. Initial rule bytes are independent uniform draws from 0–255, initial phenotype bits are independent fair binary draws, and both layers have fixed zero boundaries. Each genotype update uses the neighbour blend and one uniformly sampled elementary write described in Box 2. A layer’s change rate is the mean fraction of cells differing between consecutive rows over the final 40*

generations, averaged over the three seeds. The layer is alive when this rate is at least 0.10 and dead otherwise. Scoring genotype and phenotype separately gives four regimes: live/live, live/dead, dead/live, and dead/dead.

The trend becomes an exhaustive finite-run observation once every operator is tabulated. We ran all 40,320 permutation-propagators and grouped them by cycle type (Table 1).

Empirical finding 1: The full permutation census

Every permutation containing an odd cycle lacks a fixed byte (Theorem 1); under the census protocol, all 29,295 such operators were classified genotype-alive. The theorem excludes a common fixed byte for all elementary writes; the observed 100% alive rate is an additional dynamical result. Every all-even cycle type splits between live and dead operators. Among those types, the alive fraction falls as the number of cycles k increases: $[8]$ ($k=1$) is 83% alive, $[2\ 6]$ and $[4\ 4]$ ($k=2$) 70/67%, $[2\ 2\ 4]$ ($k=3$) 49%, and $[2\ 2\ 2\ 2]$ ($k=4$) 26%. Overall, 92.5% have a live genotype. The regime split is 82.7% live/live, 9.8% live/dead, 6.0% dead/dead, and 1.4% dead/live, so all four regimes occur among bijective operators.

Table 1

Genotype-alive fraction by cycle type under the census protocol, over all 40,320 permutations. Any odd cycle forbids a fixed byte (Theorem 1); empirically, every such operator is alive. The all-even types split, with collapse becoming more frequent as the number of cycles k increases.

cycle type	count	genotype-alive
any type with an odd cycle (17 types)	29,295	100.0%
[8]	5,040	82.9%
[2 6]	3,360	70.0%
[4 4]	1,260	66.7%
[2 2 4]	1,260	49.2%
[2 2 2 2]	105	25.7%

Fixed Points Are Not Attractors

Worked example 4: Reading the rotation survey

The fields of Figure 6 evolve under the *spatially coupled* dynamics of *Fixed Points Are Not Attractors*—stochastic elementary writes together with the coupling between cells—not P_σ applied cell-by-cell. This is what the survey turns on. Without the blend, Figure 5 already shows rotate+1 remaining active over the observed horizon while rotate+2 settles. With neighbour blending, the finite-horizon split persists and widens: the single 8-cycles— $\gcd(\text{offset}, 8) = 1$, offset ± 1 and ± 3 —sustain a high-diversity field, while offset ± 2 ($\gcd 2$) and ± 4 ($\gcd 4$) collapse; that split is what the survey shows.

The two panels are *almost* mirror images, and where they differ is instructive. Offset $+d$ and offset $-d$ are the same permutation exactly when $2d \equiv 0 \pmod{8}$ —so offset ± 4 (and the identity) are one operator with bit-identical panels, whereas offset ± 1 and ± 3 are distinct *inverse* rotations. Inverses share cycle structure, fixed rules, and the sustain-or-collapse verdict, but not their transient dynamics: offset $+1$ and offset -1 realise the same sustaining regime through visibly different fields.

Possessing a fixed point (Theorem 1) is not the same as reaching one, and P_σ does not by itself move a rule toward $\lambda = 1/2$. Because P_σ complements every bit, $\lambda(P_\sigma g) = 1 - \lambda(g)$: the operator *reflects* activity across $1/2$, exactly preserving the distance $|\lambda - 1/2|$ rather than contracting toward it. And because P_σ is a bijection on bytes, a generic rule lies on a periodic orbit and simply cycles—its fixed points are fixed, but they are not attractors and have no basins. The only rules pinned at $\lambda = 1/2$ are the fixed points themselves. Attraction in the experiments therefore belongs to the stochastic elementary-write dynamics, with or without neighbour coupling, rather than to iteration of the simultaneous map P_σ .

Empirical finding 2: The rotation survey

Within the wrap-rotation subfamily, every non-trivial rotation is all-even and so possesses fixed rules (Theorem 1), yet under the neighbour-blend coupling the tested rotations split by cycle structure (Figures 6, 7):

- the single 8-cycles— $\gcd(\text{offset}, 8) = 1$, offset ± 1 and ± 3 —*sustain* a high-diversity field, holding ~ 30 – 40 distinct rules through the longest runs we made (400 generations);
- offset ± 2 ($\gcd 2$, two 4-cycles) and offset ± 4 ($\gcd 4$, four 2-cycles) *collapse* to one or two rules within a few dozen generations.

The distinct-rule population curve (Figure 7) separates the two by more than an order of magnitude in a single scalar. Possessing a fixed rule and having the coupled field stay near it are different properties. The present claim is limited to two seeds per rotation and treats population size as a descriptive observable, not as an order parameter in the statistical-mechanics sense.

Constructive control by temporal schedules

Definition 5 (Temporal braid). *A temporal braid alternates two writings between complete coupled spatial updates: writing w_A is used at even times and w_B at odd times. More generally, a periodic schedule selects one writing at each time step. This preserves the intermediate spatial states and neighbour blends; it is not the algebraic composition $P_{\sigma_A} \circ P_{\sigma_B}$ of two simultaneous maps.*

Empirical finding 3: A braid of two collapsing operators can sustain the field

At width 80, horizon 120, and seed 0, offset +2 and offset +4 each have genotype change rate 0 when held alone, while their alternating braid has change rate 0.76 (Figure 9, top). The matched schedule control offset +2/offset −2 remains dead (Figure 9, bottom). This establishes that two collapsing components can construct a sustaining schedule. It does not establish the previously suggested block-system criterion: all three even offsets preserve the evens/odds partition, and both pairs generate the same translation subgroup $\langle +2 \rangle$. The cause of the contrasting braid outcomes remains to be identified.

Empirical finding 4: The mixing fraction is a reproducible diversity control

Let σ be displayed-position offset +1. We ran periodic 12-step schedules mixing the sustaining 8-cycle σ^1 with the collapsing two-4-cycle operator σ^2 , at width 80 for 120 generations and five seeds per mixture. Mean late-window diversity rises from 1.0 rule when the σ^1 fraction is zero, through 16.8 at fraction 0.25, to 36.9 at fraction 5/12; above that it remains in the range 34–41 rather than increasing monotonically (Figure 10). The schedule fraction is thus a tunable control for entry into a sustained high-diversity regime, with saturation rather than a continuously linear response.

Definition 6 (Interrupter). *An interrupter is an update between elementary propagator writes that is not itself a propagator write. The coordinate-wise neighbour blend of Box 2*

is the interrupter in the ordinary spatial runs; a hold gate that suppresses a scheduled write is another example. An interrupter changes the coupled dynamics but does not, by definition, guarantee sustained diversity.

Empirical finding 5: The rotation split occurs with and without neighbour blending

Figure 5 uses stochastic elementary writes without neighbour blending; Figures 6 and 7 include the blend. Over the displayed horizons, offset +1 remains active and offset +2 settles under both protocols, although their diversity levels and transients differ. These runs therefore do not identify neighbour blending as the cause of the split; they show that fixed-byte existence alone does not determine finite-horizon settling under either protocol.

Proposition 1 (Composition parity). *Products alternate between complemented and uncomplemented coordinate permutations. Each operator of Equation (1) negates every bit ($P_\sigma(g)[j] = \neg g[\sigma^{-1}(j)]$, a coordinate permutation followed by a global flip), so under composition the flips cancel in pairs: $P_\sigma \circ P_\tau$ is a pure coordinate permutation with no negation left—a bijection that fixes 2^c bytes for c cycles but carries none of the forced $\lambda = 1/2$ structure. An even number of propagators therefore composes to an uncomplemented coordinate permutation, and an odd number to another member of the permutation-propagator family.*

Exploratory searches over products of two and three simultaneous propagators found no additional structural regime. Temporal braiding is a different operation: it alternates stochastic elementary writes inside the spatially coupled dynamics rather than replacing them by one composed map.

Algebraic composition is therefore closed in the sense of Proposition 1. Temporal scheduling is different: it retains each intermediate coupled state, which is why the braids

of *Constructive control by temporal schedules* can produce dynamics absent from either held operator. A second constructive mechanism pairs a propagator with a step of a different kind.

Definition 7 (River). *A river is the composed operator obtained by following the collapsing offset-+2 propagator ($\text{gcd } 2$) with a phenotype-reading construction step—match the local phenotype against four template candidates, first match wins, and fall back to the all-zero rule when none matches.*

Empirical finding 6: The river sustains structured phenotype

In all six tested seeds at $L = 80$, $T = 120$, the river produces coherent bands of alternating phenotype that drift through a disordered background. In the larger pinned run at $L = 240$, $T = 360$ (Figure 11), those structures persist while branching, merging, and reorganising across the full spacetime field. This is a bounded six-seed observation plus one larger representative run, not a claim that the structure is an attractor or survives arbitrarily long.

Limits of the Tested Distance and Complexity Measures

Empirical finding 7: The tested per-trajectory metrics do not separate the parity classes

The classification is a conjugacy invariant of σ —its cycle type—while the complexity and compression measures we tried read a single rule’s bits or one space–time trajectory without explicitly encoding σ . In our tests these measures did not recover the parity classification: a population count reads a collapsed-but-maximally-selected field as nearly dead; a frozen periodic genotype scores a lower statistical complexity C_μ (Crutchfield & Young, 1989) than Rule 110; gzip rates a self-similar rule as noise. This is a negative result about the tested summaries, not an impossibility theorem for trajectory-based inference.

Discussion

The survey (Figure 6) exposes a sharp behavioural split within a family of provably fixed-point-bearing operators: only the single 8-cycles (offset ± 1 , ± 3) sustain a high-diversity phase, while offset ± 2 and ± 4 collapse. A reader is entitled to ask whether that phase sits at a *critical point*—tuned between an ordered and a disordered phase, as CA complexity is often supposed to (Langton, 1990). We test this with a finite-size scan and then ask whether the sustained diversity is spatially organised into coexisting dynamical regimes.

Empirical finding 8: No finite-size evidence of a critical point over the tested range

The natural control parameter is the *propagator fraction* q ($q = 0$ all blend, $q = 1$ all propagator, offset +2). Across widths $L = 30\text{--}240$ (32 seeds each) the genotype activity $a_G(q)$ rises *smoothly*, well described by a logistic crossover ($R^2 > 0.99$, Figure 12a); its width does not shrink over the tested sizes ($w \approx 0.13$, $w \sim L^{-0.04}$). The size-scaled run-to-run spread $L \text{Var}(a_G)$ (Figure 12b), used as a finite-size susceptibility proxy, *peaks* inside the crossover band, but the peak drifts in location ($q = 0.75, 0.50, 1.00, 1.00$ as L increases) and grows $\sim 5.7\times$ from the smallest width to the largest; for the two largest sizes its maximum lies at the sampled boundary $q = 1$, so no convergent interior peak is resolved. Together with the non-shrinking logistic width and absence of a Binder-cumulant crossing, the scan provides no finite-size evidence for a sharpening transition over $L = 30\text{--}240$. These results support a broad crossover, plausibly influenced by seed-dependent residence times, rather than a confirmed critical point; they do not exclude different scaling outside the tested range.

Empirical finding 9: A finite-width intermediate-activity band

An intermediate-activity band ($0.15 \leq a_G \leq 0.60$) has a *lower* edge robustly at $q \approx 0.1$ —stable under both system size and threshold—while its upper edge is threshold-dependent, because a_G saturates gradually rather than jumping (the band spans $q \approx 0.1$ to 0.4 – 0.5). This finite-width crossover is what we call a *littoral edge*: a descriptive name for a shore of finite width, not a thermodynamic phase boundary.

Definition 8 (Isolated-rule activity score). *The activity score of a rule is the mean per-step cell-change fraction over the final 150 steps of a 300-step ordinary ECA run on a periodic line of width 300. Every rule uses the same density-1/2 pseudorandom initial condition (seed 1). This finite-run measure is distinct from the static truth-table balance λ (Every Fixed Point Is Balanced). Ordered rules score near 0; Rule 110 scores approximately 0.42, and several standard chaotic rules score approximately 0.50. The score orders activity levels but does not by itself distinguish complex from chaotic dynamics.*

Tinting the genotype by this score (Figure 13) resolves the intermediate field into coexisting low- and high-activity domains whose boundaries trace filamentary sets through 1+1-dimensional spacetime—objects *analogous to* the domain boundaries of computational mechanics (Crutchfield & Young, 1989; Shalizi & Crutchfield, 2001), here obtained by thresholding a scalar tint rather than by predictive equivalence.

Empirical finding 10: Filamentary walls have elevated genotype churn

The activity field resolves into ordered and chaotic domains separated by *fractal-like walls* (Figure 14; box-counting dimension $D \approx 1.5$ – 1.7 over $L = 128$ – 768 , a handful of seeds). Those walls are the locus of elevated *genotypic* change: measuring the row-to-row rewrite rate by region (Table 2), the wall churn exceeds the chaotic interior in all 32 committed operator/width/seed fields and exceeds *both* interiors in 24. At $L = 256$, offset +1 gives 0.80 at the wall against 0.81 (ordered) and 0.73 (chaotic);

Table 2

Genotype churn by region—row-to-row rewrite rate of the rule field (fraction of cells whose ECA rule changes from the row above), at $L = 256$, mean over the committed seeds. The wall exceeds the chaotic interior for every operator; for offset +1, the ordered interior is slightly higher. Wall extraction is the threshold-and-smooth procedure of Figure 14; reproducible from the generated fields (`plot_eoc_churn.py`).

Operator	Wall	Ordered interior	Chaotic interior
offset +1	0.80	0.81	0.73
$\sigma = 16250374$	0.71	0.68	0.57
river	0.99	0.97	0.96

the corresponding wall is the maximum region for $\sigma = 16250374$ and the river. The chaotic domain has the lowest aggregate churn for every operator and width (and in 29 of the 32 individual fields), so the wall—where ordered abuts chaotic—is consistently more rewritten than the chaotic interior, though not universally more than the ordered one. The walls are not compositionally special (no rule class enriched; Rule 110 is, if anything, depleted), so the distinction is dynamical rather than a difference in rule composition. The interface analysis is exploratory: it uses a handful of seeds and a fixed-boundary variant, and the denser river boundary ($D \approx 1.8$ versus 1.5–1.7) is an unmatched comparison that does not establish an effect of genotype–phenotype coupling. Whether rewriting transports or combines information *along* the walls is taken up in *A Sampled Edge of Chaos*.

A Sampled Edge of Chaos

If the filaments are domain boundaries they are candidate *wires*. The natural question is not which rules compose them—in our data the complex rules are not enriched there—but whether information is *transported along* them and *combined where they collide*

(Mitchell et al., 1993). We answer both within one synthetic family, and are explicit about what that buys and what it does not.

We read both notions off the binary activity field as local transfer entropy (Lizier et al., 2008): the apparent transfer entropy from a spatial neighbour into a cell, conditioned on the cell’s own past, scores directed *transport*; the *synergy* of the two neighbours—their joint predictive contribution beyond the sum of their separate ones—scores *combining* (Figure 15).

A raw on-filament average will not support either claim, and we say why before using one. The score has two failure modes, and each needs its own null. First, the plug-in estimator is positively biased, so a field can score positive transfer entropy with no directed coupling at all; the null for this is a *source time-shuffle*, which preserves the marginals and the cell’s own past and destroys only the neighbour-to-target coupling. Second—and this is what defeats the raw statistic—the extracted filament is not a fixed object across a parameter sweep. It swells from 20% of the field at the bare operator to 39% under strong noise (Figure 16c), at which point it is no longer a wall, and any on-filament mean drifts toward the field mean for reasons that have nothing to do with transport. The null for this is a *mask-matched* draw: the same number of cells, taken anywhere in the field. We report the difference from that draw and call it *filament-specific* transport or combining; it is zero when the filament is no better than an arbitrary set of cells of the same size, and negative when it is worse.

A size-matched draw preserves the filament’s extent but not its shape, so we also ran two nulls that preserve its geometry. The first circularly translates the mask along x and averages over all displacements beyond the point where a mask ceases to overlap itself; this retains the mask’s exact connectivity, anisotropy, cell count and per-row occupancy, destroying only its registration against the field, and it involves no randomness at all. The second scores each field under the filament extracted from an independent seed at the same sweep point. Both agree with the size-matched draw to within 0.001 bits at every point

($r > 0.999$ across the sweep), and every result below is unchanged under all three. A filament-shaped region placed generically therefore scores no differently from scattered cells: what distinguishes the real filament is where it lies, not the shape of the region measured. We quote the size-matched null throughout.

The estimator has a third failure mode that no null removes, and it bounds what follows. Where the activity field is nearly degenerate—almost all cells in one state—the conditional probabilities driving the score become extreme and it inflates without bound, and this happens at *either* end. The genotypically dead offset +4 operator is the demonstration. Its genotype freezes (row-to-row rewrite rate falls to 0, Empirical finding 3) into a population that is 98% active, and on that saturated field it records the largest on-filament transport we measured anywhere: $317\times$ the domain interior, +0.59 bits filament-specific, for an operator that transports nothing at all. The sparse end behaves the same way, and costs us the most ordered sample in the sweep below (all blend, $q = 0$, 9% active), where the shuffled surrogate outscores the real field (+0.144 against +0.105 bits, $t = -0.71$, $p = 0.49$); that point is bias-dominated and is excluded. Local transfer entropy is therefore comparable only across fields of similar and non-degenerate activity density—which is the concrete content of what earlier drafts recorded as a “sparsity effect”.

Within these bounds the bare operator’s filaments transport. Over the eight seeds of the $q = 1$ sweep column, on-filament transfer entropy is $+0.158 \pm 0.005$ bits against a shuffled surrogate of +0.002 ($t = 33.5$, $p < 10^{-8}$) and a mask-matched draw of +0.061, leaving $+0.097 \pm 0.005$ bits of filament-specific transport. On the single fixed-boundary field of Figure 14 the same measure concentrates $5.7\times$ over the domain interiors while its surrogate collapses to +0.0002 bits, leaving +0.135 bits filament-specific; the Rule 110-dominated $\sigma = 16250374$ field, as a positive control, shows the same signature ($7.6\times$, surrogate +0.001, filament-specific +0.161 bits).

To ask whether combining is a property of the *edge* rather than of the operator alone, we sample a synthetic order–chaos family. The neighbour-blend interrupter (*Fixed*

Points Are Not Attractors, fraction q) tunes toward order; an independent, seeded stochastic rule-flip applied to the genotype each step (rate p) tunes toward chaos; the bare propagator sits between them at $q = 1, p = 0$. Figure 16 reports both filament-specific measures across this family.

Observation 3: Transport and combining are elevated together in a transitional band

Averaged over eight seeds per point, both filament-specific measures rise out of the ordered regime, are elevated across an intermediate band of blend fractions ($q \approx 0.3$ – 0.7 ; transport $+0.155$ bits, combining $+0.041$ over the band), and decay monotonically through the bare propagator into the noise-driven regime (Spearman $\rho = -0.97$ and -0.99 over the chaotic flank, $n = 64$). The band exceeds the bare operator on both measures ($t = 8.8, p < 10^{-8}$; $t = 11.0, p < 10^{-7}$) and exceeds the ordered samples at $q \leq 0.2$ ($p < 0.001$). Its interior is not resolved: the largest single value falls at $q = 0.4$ (transport $+0.172 \pm 0.011$ bits, combining $+0.0444 \pm 0.0012$), but no other point in the band separates from it on transport ($p \geq 0.058$), so we report a band and not a peak. Combining is thus largest in a transitional regime—a literal edge of chaos in the sense of Langton (1990)—and is suppressed by both over-ordering and disorder; the bare offset+1 operator sits below the band, on its descending side.

The two measures part company under noise, and the split is the sharper result. Filament-specific transport falls to zero and then *inverts*: it is positive at $p = 0.2$ ($+0.012, p < 10^{-5}$) and significantly negative from $p = 0.4$ onward ($-0.005, p < 10^{-4}$; -0.012 at $p = 0.7$), meaning the extracted wall carries less directed transport than an arbitrary set of cells of the same size. Directed coupling itself does not vanish there—the raw score still stands far above its shuffled surrogate—so noise does not destroy transport in the field; it destroys the filament’s ownership of it. Combining remains weakly filament-specific throughout ($+0.0028$ at $p = 1, p < 10^{-8}$).

Two things bound this. The ascent into the band is not a density artefact: across $q = 0.1\text{--}0.4$, where filament-specific transport grows 3.7-fold, the filament fraction is flat (0.086–0.089) and activity density rises, which is the direction that would *depress* the score given the sparse-limit inflation above. The descent is weaker, because there the filament fraction climbs monotonically alongside the decay and the two cannot be separated by these data; what survives on that side is the sign change, which none of the three nulls can manufacture. Beyond that, the sweep is one synthetic family measured by one estimator at history length one, and a longer history or a different decomposition of the joint contribution could move the numbers.

Within those bounds both halves of the question have an answer here. Information is transported along the littoral filaments of these diagrams and combined on them, most strongly at an intermediate regime that neither the ordered nor the noise-driven extreme reaches; and where noise dissolves the walls, transport stops being theirs to carry. What we cannot yet say is whether this holds beyond the single family sampled here—that is the question for another paper.

Domains and Particles

The structures reported above have an ancestor. Computational mechanics reads a cellular-automaton spacetime diagram not as one pattern but as a decomposition: *domains*, the regular backgrounds a rule settles into; *particles*, the localised structures that separate one domain from another and propagate through the diagram; and *collisions*, the interactions between particles, where the system’s computation is usually held to be taking place (Crutchfield & Hanson, 1993; Hanson & Crutchfield, 1997). Rule 110’s gliders are the canonical case. *The Object* places this paper’s *object* among cellular-automaton variants; this section places its *structures* among ways of reading them.

The correspondence runs at all three levels. Tinting the genotype by activity resolves the intermediate field into coexisting low- and high-activity regions (Figure 13): domains, in the sense above, though not obtained in the same way. The filamentary walls

separating them are localised, persistent, and propagate through spacetime with box-counting dimension 1.5–1.7 (Empirical finding 10); they occupy the position particles occupy, and they are where genotype rewriting is fastest. The river (Empirical finding 6) produces coherent bands that drift through a disordered background and then branch, merge, and reorganise—which is what a population of interacting particles looks like. That last resemblance is qualitative and we do not press it: the river’s denser boundary is an unmatched comparison and establishes nothing about genotype–phenotype coupling.

Where this departs from the ancestor is the reason the walls move. In the classical setting one fixed rule runs everywhere; a domain is a pattern in the phenotype, and a particle is a defect in that pattern, carried along by the single rule that generates both. Here the domains are regions of *different rules*, and a wall is mobile because the genotype is being rewritten underneath it. Their particles move through a static rule; ours move because the rule itself is being edited at the boundary. The two pictures agree about where to look and disagree about what is doing the moving.

We inherit the vocabulary, not the machinery. Computational mechanics obtains its domains by predictive equivalence, building the ϵ -machine of the spatial configuration and reading regularity off it (Crutchfield & Young, 1989; Shalizi & Crutchfield, 2001); ours come from thresholding a scalar activity tint, which is cheaper, coarser, and carries the caveat recorded with it—the tint is a statistic of rule space, so equal-tint neighbourhoods need not behave alike. The mapping above is therefore a reading aid rather than a derivation. A properly computational-mechanical treatment of this substrate—domains defined by predictive equivalence over the joint genotype–phenotype configuration, particles as their boundaries—would settle in a principled way what *A Sampled Edge of Chaos* approaches with a single-estimator proxy, and is the natural next instrument rather than something attempted here.

Limits

The theorems are the Boolean case and are exact; the dynamical claims are narrower and empirical. The census and the sustained-diversity behaviour are reported for a single spatial coupling (S_8 , radius one, one value map) at limited seed counts, and invite a fuller treatment. *Discussion* takes up the local, structural reading and reports an exploratory interface-dimension statistic under its own stated caveats (few seeds, fixed-boundary variant); beyond it we report no perturbation-spreading or entropy-rate statistics. The information measures of *A Sampled Edge of Chaos* are reported against three stated nulls—one size-matched, two geometry-preserving, in agreement—and within a stated density range, for one synthetic family at one history length; the local transfer entropy estimator is shown there to be unusable on near-degenerate activity fields, and we make no cross-operator claim from it. The empirical components use different fixed sample sizes: three seeds per operator in the census, two per rotation in the survey, six for the river and Rule 110 comparison, and 32 per parameter cell in the finite-size scan; the interface analysis uses the smaller samples stated with that result. Illustrative diagrams use pinned representative seeds. All are reproducible from the accompanying package. The substrate the object sits in is not itself new (*The Object*); λ is one heuristic among several (Wuensche, 1999), and we make no claim that balance implies any particular dynamics—only that here balance is *derived*. What is new, and what we rest the paper on, is the object of Equation (1) and the fixed-point structure that forces every fixed rule to be balanced, $\lambda = 1/2$, as a matter of proof.

Conclusion

The characterisation promised at the outset (*The Object*) has two halves. The first is static and exact: which operators possess fixed points (the all-even ones), how many (2^k), and where they lie (balanced, $\lambda = 1/2$, always). The second is empirical and tentative. Because every fixed point sits at $\lambda = 1/2$, that parameter is silent on the difference between an operator whose field stays diverse and one whose field collapses; the

distinct-rule count is not, and in our runs it separates the two across the rotation family by more than an order of magnitude (Figure 7). The same observable offers a suggestive reading of the search that produced the family, reported both as a representative run and as a fixed six-seed sample (*Limits*). The operator of Figure 2 that comes to be dominated by Rule 110—a conventional target in a search for computationally distinguished behaviour—concentrates the field onto it and pays for it in diversity. At the representative seed, 39% of its genotype cells are Rule 110 and it sustains 16 distinct rules, against a foil that holds Rule 110 below 1% and sustains 33; over the six-seed sample the ordering is unchanged—Rule 110 35% (range 31–39%) versus 0.2% (0–0.2%), sustained diversity 15 versus 23. On this pair, recovering a universal rule and sustaining a diverse population pull against each other—a suggestion worth testing at scale, not a law we have established.

Data and code availability. Every numerical result in this paper is regenerated from source rather than deposited: the generators are seeded, so a single documented command rebuilds the fields underlying the tables, figures, and reported summary statistics from an empty data directory, and the analysis scripts consume no input that command does not produce. The code and those commands are publicly available at <https://github.com/tothedarktowercame/mmca-clj> (commit `e3e7bc59a02a229ef87bc7b7f296be0f5dc4b0c6`)—a standalone, dependency-light Clojure port of the MetaCA dynamics whose `README` gives the complete reproduction command and commands for individual figure groups. The classification and balance theorems are formally verified in Lean 4 with Mathlib at <https://github.com/tothedarktowercame/mathlib4>, branch `darktower` (commit `bc34d8cec27216aae1d3cd511a9950ab1bce3575`), in `DarkTower/Patterns/Propagator.lean`: the cycle-parity criterion as `hasAlternatingColouring_iff_cycleType_even` and the $\lambda = 1/2$ balance as `alternatingColouring_true_card_eq_four`. The same file also contains executable regressions for both hand calculations, including the displayed-position versus

numerical-neighbourhood indexing contrast.

Generative-AI disclosure. Anthropic Claude (Opus 4.8, Opus 5, and Fable 5) and OpenAI Codex (version 5.6-sol) coding agents (Anthropic, 2026a, 2026b; OpenAI, 2026) assisted in an iterative, human-supervised workflow with manuscript editing, software implementation, test construction, and adversarial checks. Exchanges with Claude Fable 5 contributed to the initial identification of the operator family reported here. The author reviewed the resulting prose and code, reran the reported computations and formal checks, and accepts full responsibility for the manuscript. Product information was accessed in July 2026; per-source access dates are given in the bibliography.

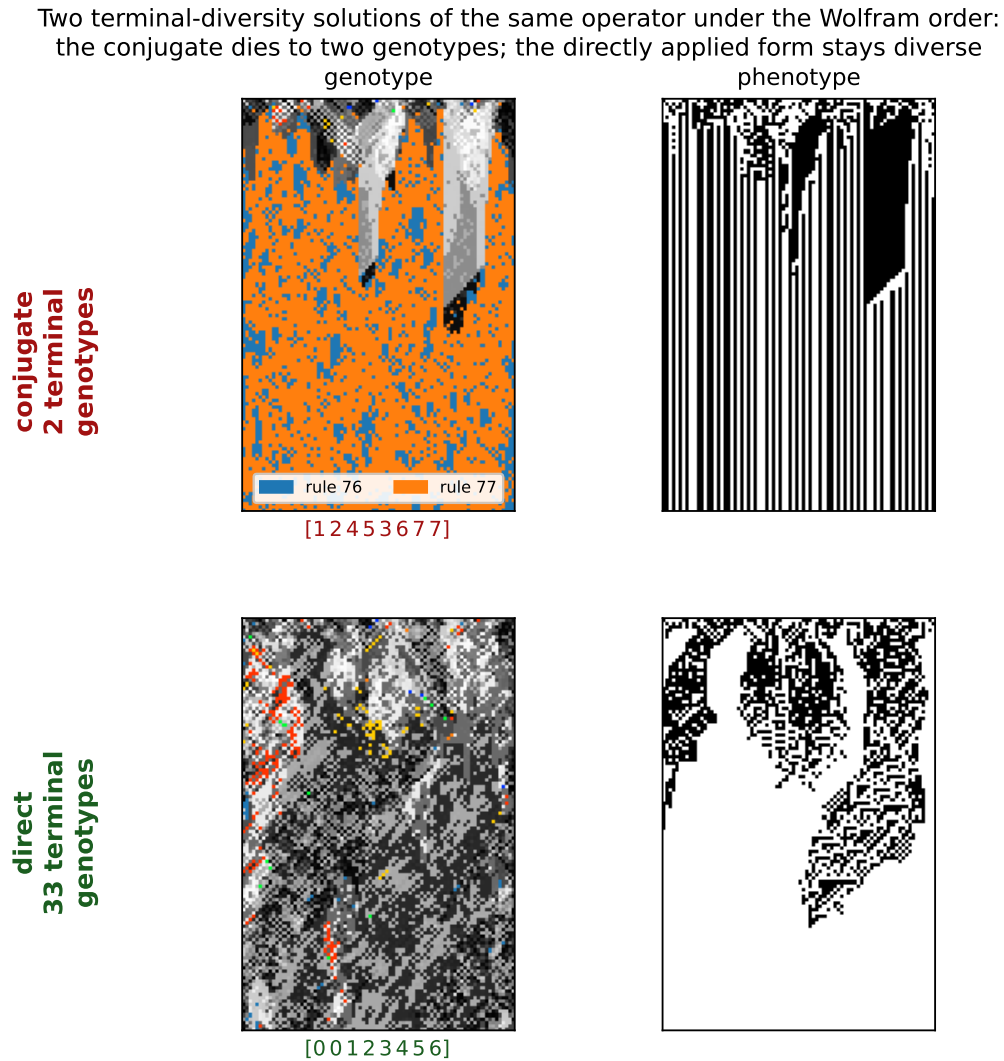
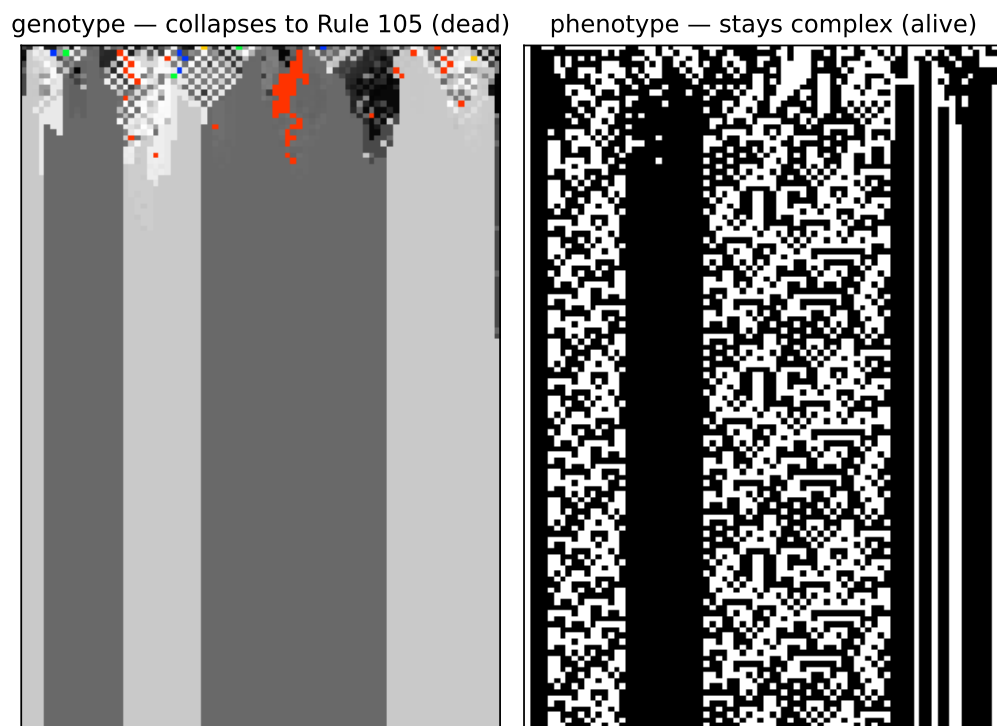


Figure 3

Two terminal-diversity solutions of one historical operator. The historical write [00123456] has two forms under the standard Wolfram order. Its conjugate [12453677] (top) reproduces the two-rule behaviour: the genotype collapses to exactly two terminal rules—76 and 77, adjacent, differing in a single bit—regardless of seed, while the phenotype stays regular. Applied directly (bottom), the same operator keeps a diverse genotype (~30 terminal rules). Two ends of genotypic aliveness from one historical write.

**Figure 4**

A genotypically dead, phenotypically live field. The all-even operator $\sigma = [23015476]$, built to fix the balanced shell containing Rule 105, collapses the genotype (left, 256-colour) to near-uniformity within a few dozen rows—two rules survive—yet the phenotype (right) never settles: the field has died onto Rule 105, an additive $\lambda = 1/2$ rule whose dynamics keep it complex. Seed 1, $W = 80$, $T = 120$.

CYCLE PARITY DICHOTOMY: a Boolean fixed point exists iff every cycle is even

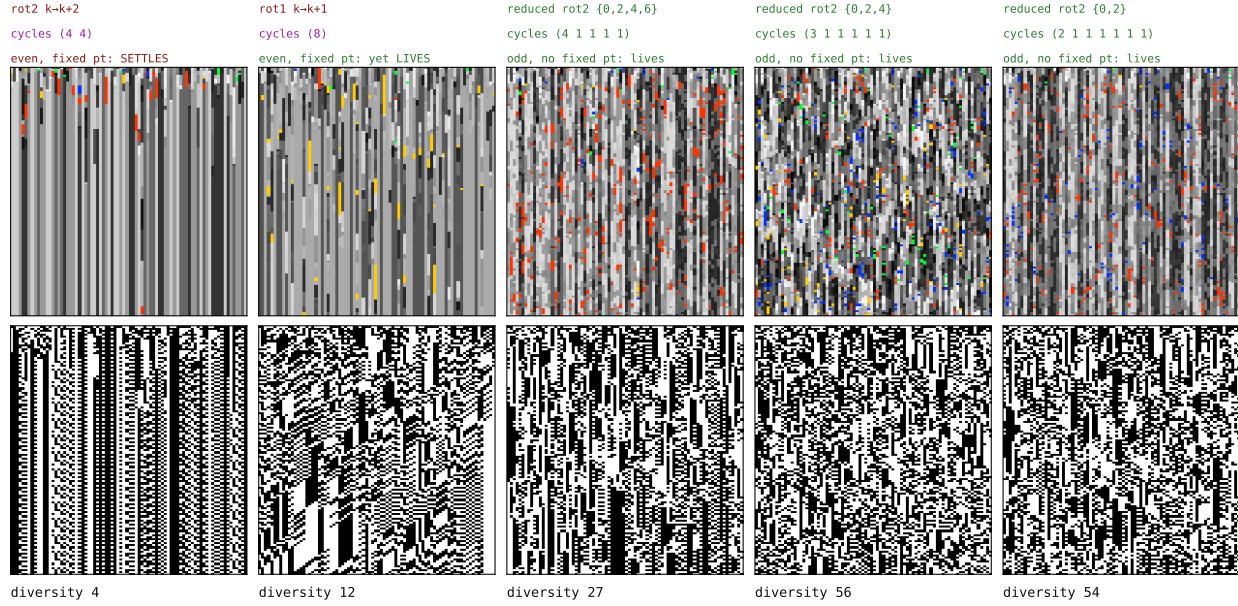


Figure 5

The parity dichotomy, by example (the census of all 40,320 permutations is summarised in the text). Five operators as genotype (top) and phenotype (bottom) over time. Left: both all-even operators possess fixed bytes (Theorem 1), but only rotate+2 settles within the run, reaching its $2^k = 2^2$ fixed rules; the single-8-cycle rotate+1 remains active at diversity 12. Right: the odd-cycle operators (reduced rotate+2, acting on a subset so the remaining positions are odd one-cycles) possess no fixed byte and stay diverse in these runs. All five use stochastic elementary writes without neighbour blending. The figure therefore illustrates the theorem's fixed-byte classification while also showing that all-even parity makes settling possible, not inevitable on a finite horizon.

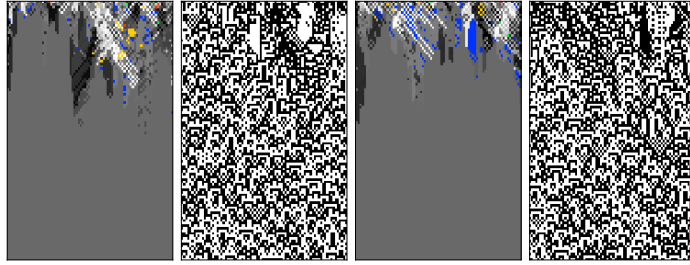
A family of operators, ranked by cycle structure (a · negative offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

offset -4

gcd 4 · four 2-cycles

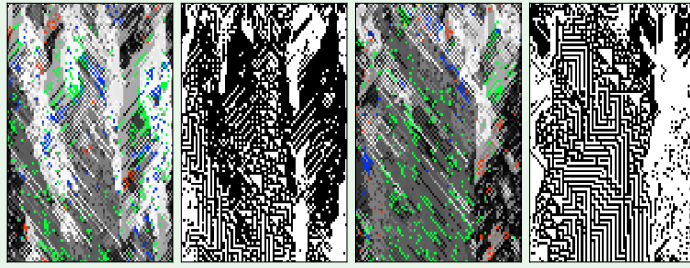
collapses ~t45



offset -3

gcd 1 · one 8-cycle

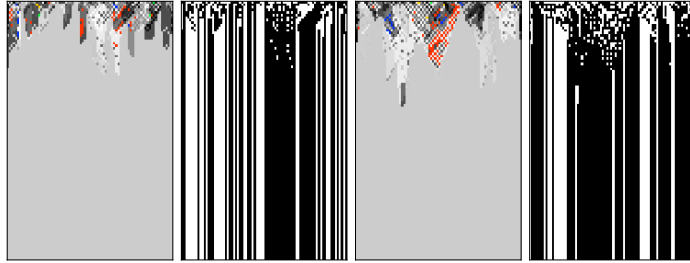
no collapse (~t100)



offset -2

gcd 2 · two 4-cycles

collapses ~t28



offset -1

gcd 1 · one 8-cycle

no collapse (~t100)

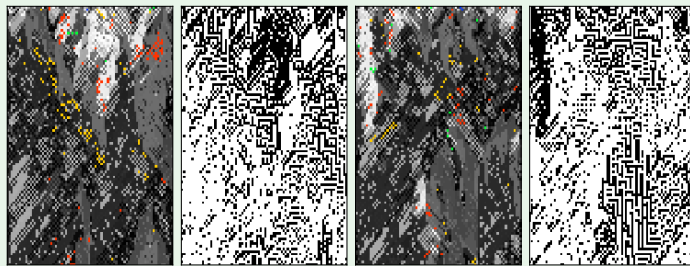


Figure 6

(a) Negative offsets. The rotation subfamily, ordered by offset: each row is one wrap rotation $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$ (two seeds; genotype in 256 colours, phenotype black and white, first hundred generations). Of these all-even rotations only the single 8-cycles, $\text{gcd}(\text{offset}, 8) = 1$ —offset ± 1 and ± 3 , highlighted—sustain a high-diversity phase; offset ± 2 (gcd 2) and ± 4 (gcd 4) collapse. See Worked Example 4; continued in (b).

A family of operators, ranked by cycle structure (b · positive offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

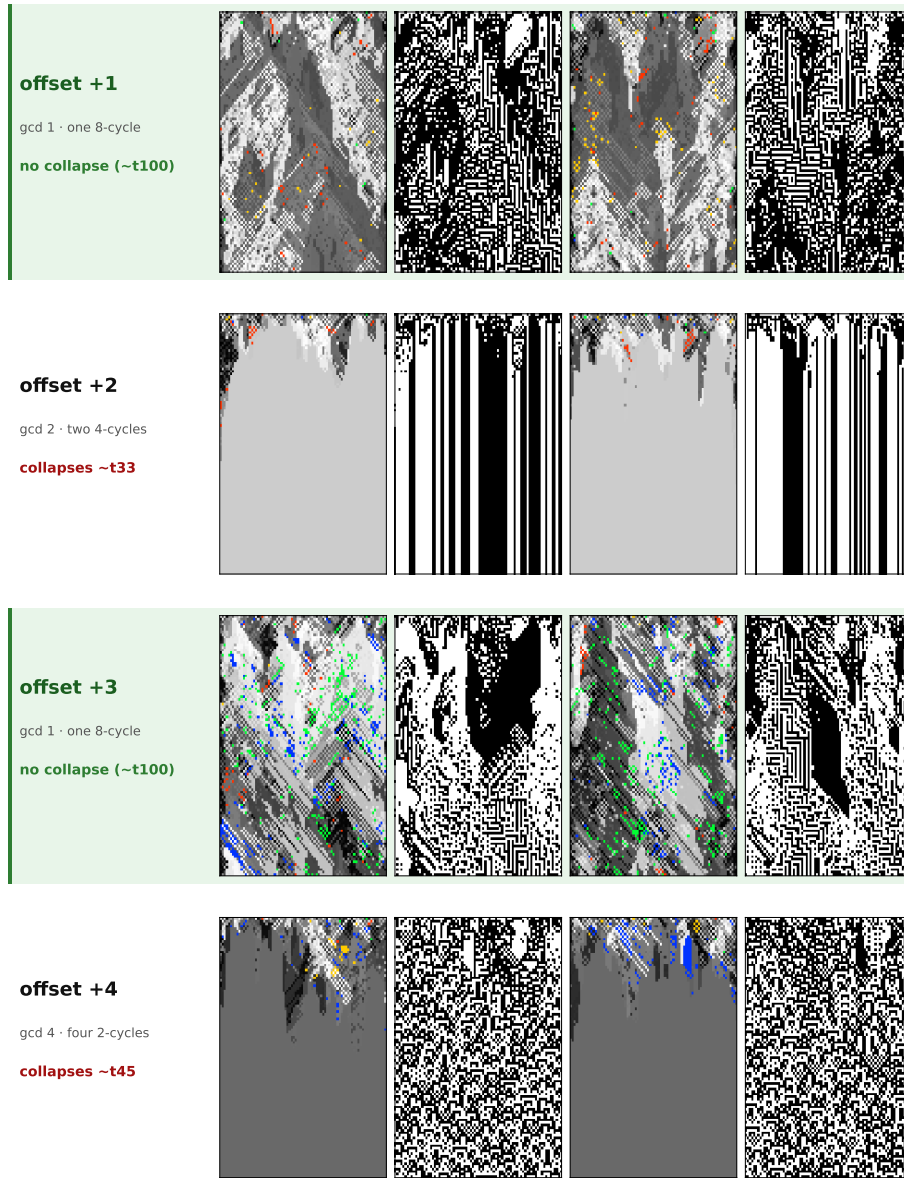
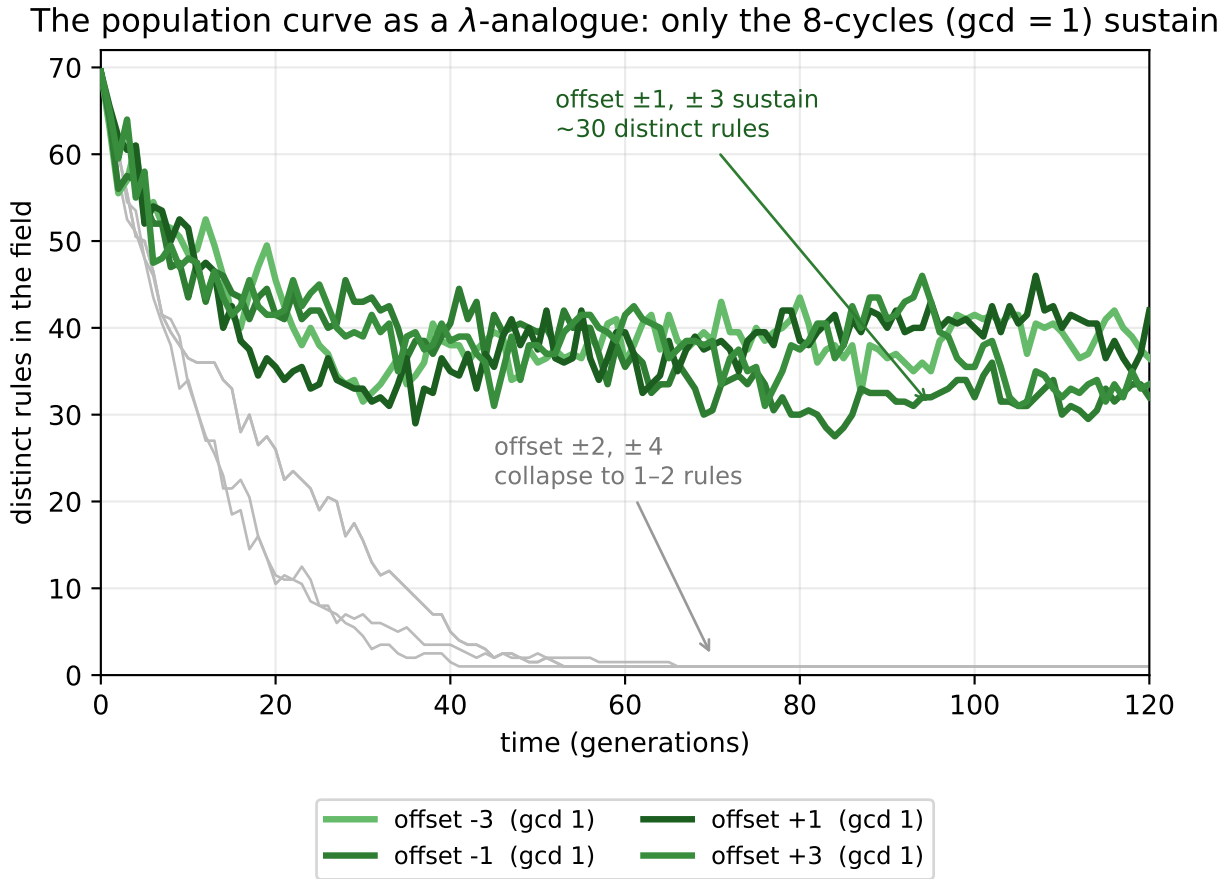


Figure 6

(b) Positive offsets. Continued from (a). The classification is symmetric under $d \rightarrow -d$ (verdict follows $\gcd(d, 8)$), so offset +1 sustains as offset -1 does, and offset +2 collapses as offset -2 does—inverse rotations sharing verdict but not their transients (Worked Example 4). Possessing a fixed point and staying near it are different properties: the two pages visually summarise the distinction between fixed-point existence and spatial persistence.

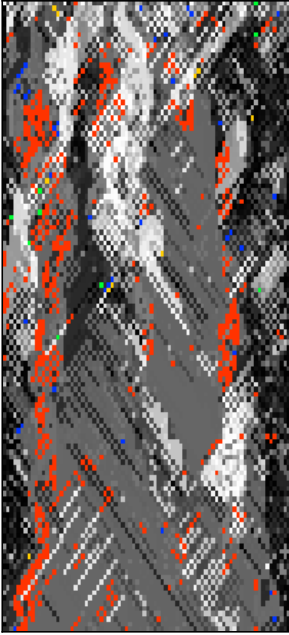
**Figure 7**

A population observable: distinct rules in the field over time, for the wrap rotations (two seeds each). All begin at ≈ 70 ; the four 8-cycles, $\gcd(\text{offset}, 8) = 1$ (offset ± 1 and ± 3 , green), keep a large diverse population, while offset ± 2 and ± 4 (grey) collapse to one or two rules. This makes quantitative, on the runs shown, the difference visible by eye in Figure 6. Figure 5 gives the corresponding without-blend comparison for offsets $+1$ and $+2$: the former remains active at lower diversity over that run, while the latter settles onto its four fixed rules.

Same cycle type (two 4-cycles), opposite fate: cycle structure does not determine aliveness

[6 7 0 2 1 4 3 5] — **two 4-cycles**

sustains (live / live)
genotype

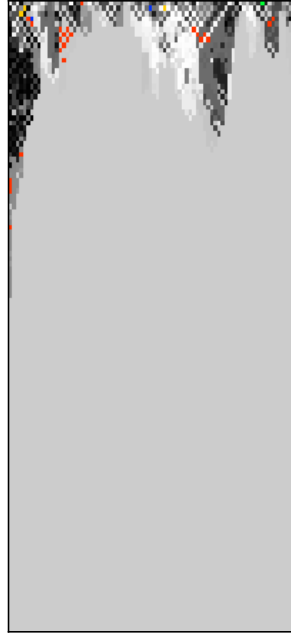


phenotype



[2 3 4 5 6 7 0 1] = **offset +2** — **two 4-cycles**

collapses (dead / dead)
genotype



phenotype



Figure 8

Cycle structure does not determine aliveness. Two operators of identical cycle type—both two 4-cycles—with opposite fate. Left, [6 7 0 2 1 4 3 5] sustains a live genotype and phenotype; it is the standard-Wolfram-order form of the historical offset+2 operator (its conjugate). Right, the Wolfram-order rotation offset+2 = [2 3 4 5 6 7 0 1] has the same two-4-cycle structure yet collapses to a dead genotype and a frozen phenotype. Same cycle type, opposite outcome: the specific permutation, not its cycle type alone, decides. A full census of the 40320 operators confirms the two-4-cycle class splits between sustaining and collapsing.

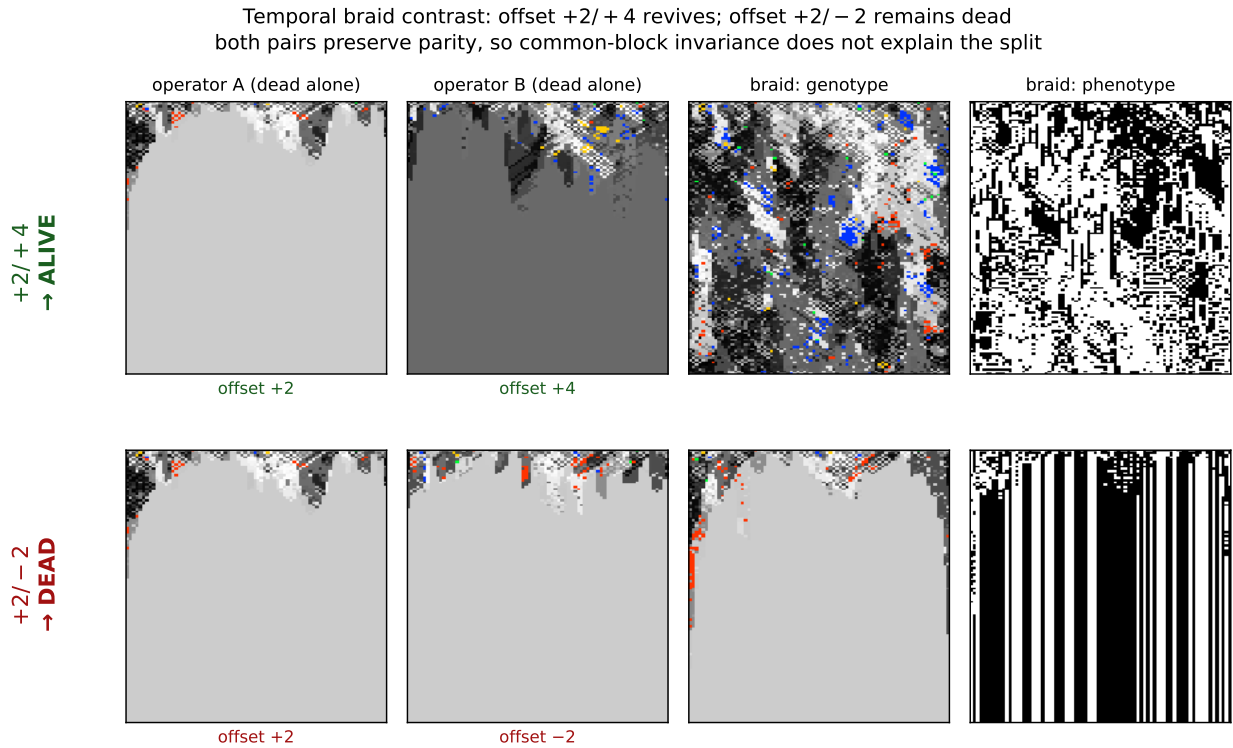


Figure 9

Temporal braiding can sustain aliveness. Two operators that each collapse alone are braided by alternating their elementary spatial updates. Top: offset+2 (two 4-cycles) and offset+4 (four 2-cycles) are each genotypically dead, yet their braid is alive (genotype change-rate 0.76, vs. 0 for either alone). Bottom: offset+2 braided with offset-2 stays dead. Both tested pairs preserve parity, so the contrast is not explained by the existence of that common invariant partition. The result concerns time-dependent coupled dynamics, not composition of the simultaneous maps P_σ .

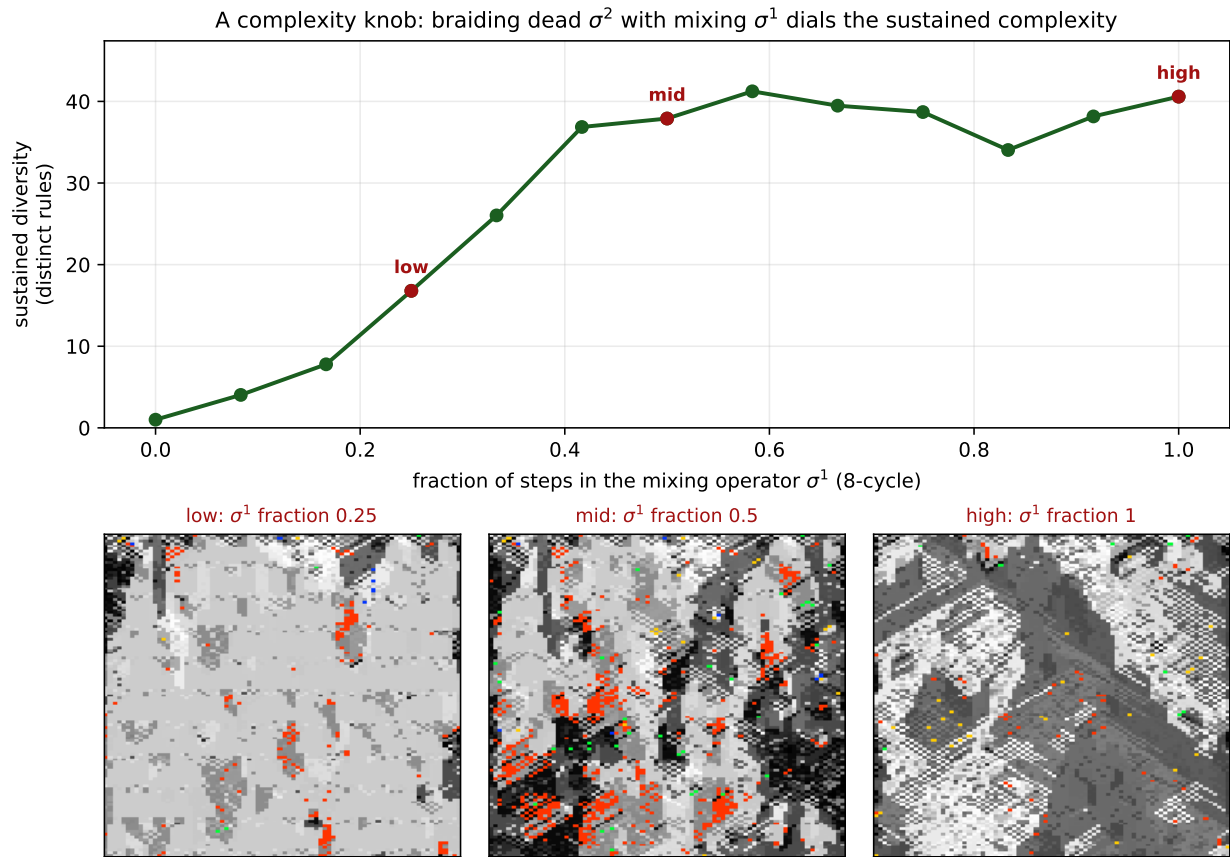


Figure 10

*A **diversity-control schedule**. Periodic 12-step schedules mix the sustaining offset+1 operator σ^1 with the collapsing offset+2 operator σ^2 . Mean late-window distinct-rule diversity (five seeds) rises from 1.0 with no σ^1 steps to a high-diversity plateau once its fraction reaches roughly 0.4. The three fields show fractions 0.25, 0.50, and 1.0 at the same representative seed.*

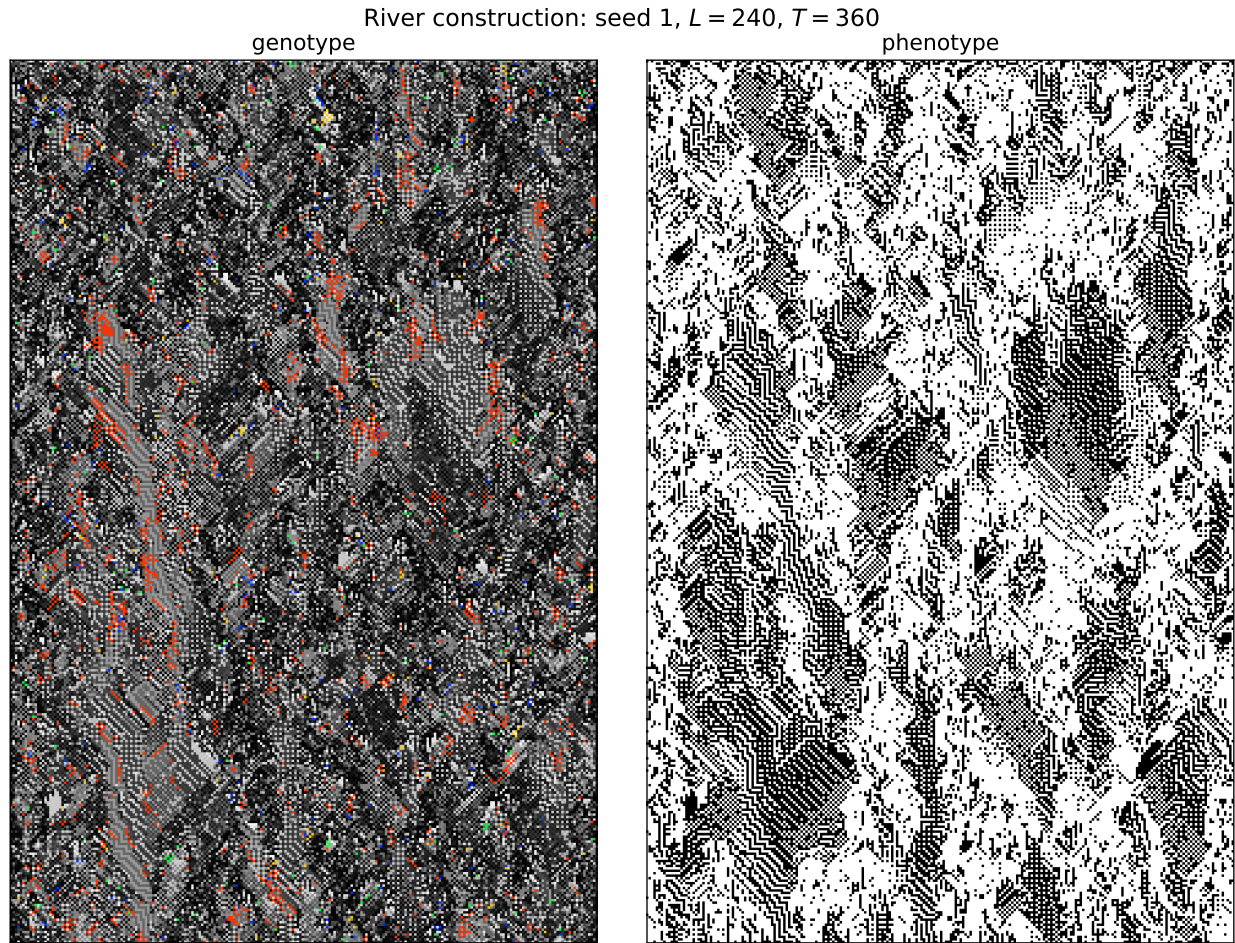


Figure 11

A larger river spacetime plate. One pinned representative run (seed 1) at $L = 240$, $T = 360$, with genotype at left and phenotype at right. The offset-+2 propagator is composed with a phenotype-reading, four-candidate, first-match template construction (fallback: the all-zero rule). Alternating phenotype domains branch, merge, and reorganise throughout the run rather than resolving to a homogeneous or short-period field. Genotype is coloured by rule as in Figures 2–6: greyscale by byte value, with well-known ECAs highlighted and red denoting Rule 110. Where the bare rotations collapse or cycle (Figure 6), this composition sustains structured phenotype for the length of the observed run.

offset+1 feedforward scan across system size L (row width in cells, 30–240; 32 seeds): a smooth crossover with no critical point, and a susceptibility that peaks in the band but drifts and barely grows—metastability, not criticality (bottom: one realization across q , $L = 240$)

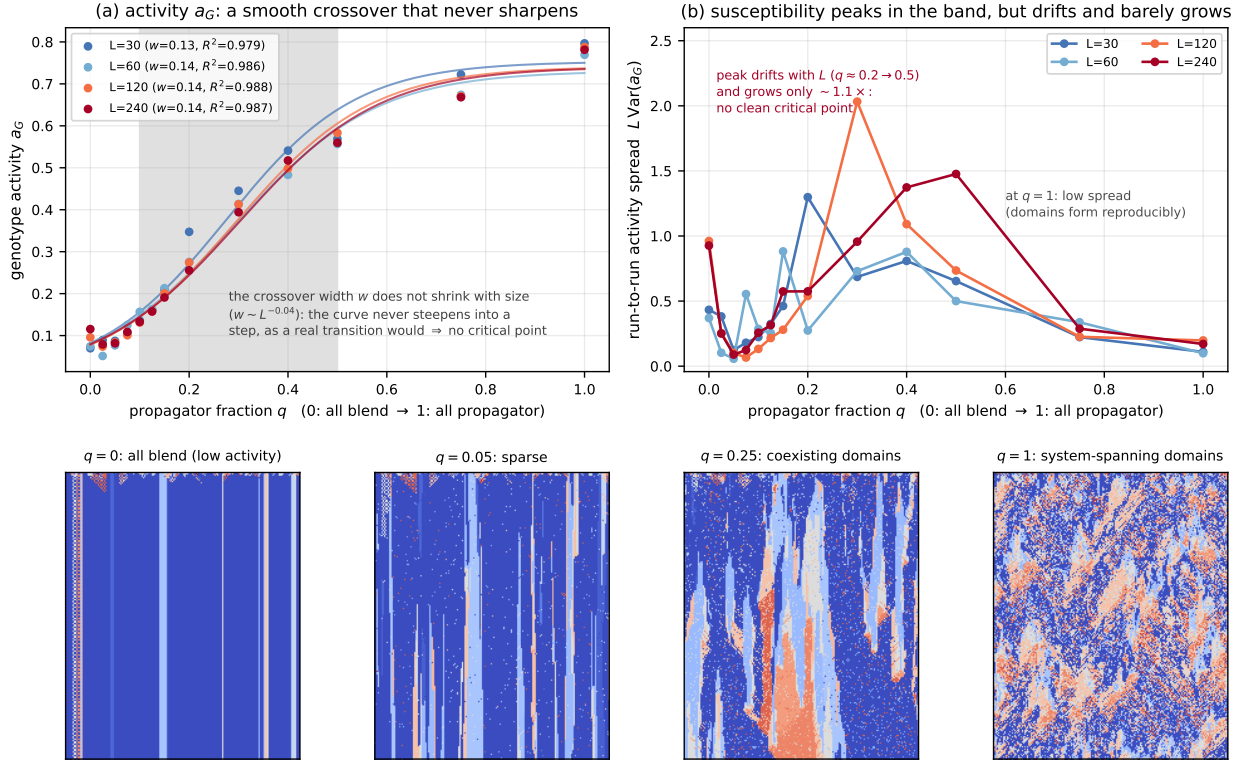


Figure 12

No finite-size evidence of a critical point over the tested range. Offset +1 feedforward finite-size scan against the propagator fraction q (0: all blend, 1: all propagator). L is the system size—the lattice width in cells; the scan spans $L = 30$ –240 (32 seeds each), because a genuine transition sharpens as L grows while a crossover does not. (a) genotype activity a_G with logistic fits ($R^2 > 0.99$); the crossover width does not shrink over these sizes ($w \sim L^{-0.04}$). (b) the size-scaled run-to-run spread $L\text{Var}(a_G)$ is used as a susceptibility proxy: it peaks inside the band, but its maximum moves from $q = 0.75$ to 0.50 and then to the sampled boundary $q = 1$ for the two largest sizes; its height grows $\sim 5.7\times$. (c) finite-horizon collapse becomes rarer with size. (d) the Binder cumulant has no common crossing. Together these diagnostics support a broad crossover rather than a confirmed critical point. Bottom: one paired realisation ($L = 240$, single seed) tinted by isolated-rule activity at $q = 0, 0.05, 0.25$, and 0.75 .

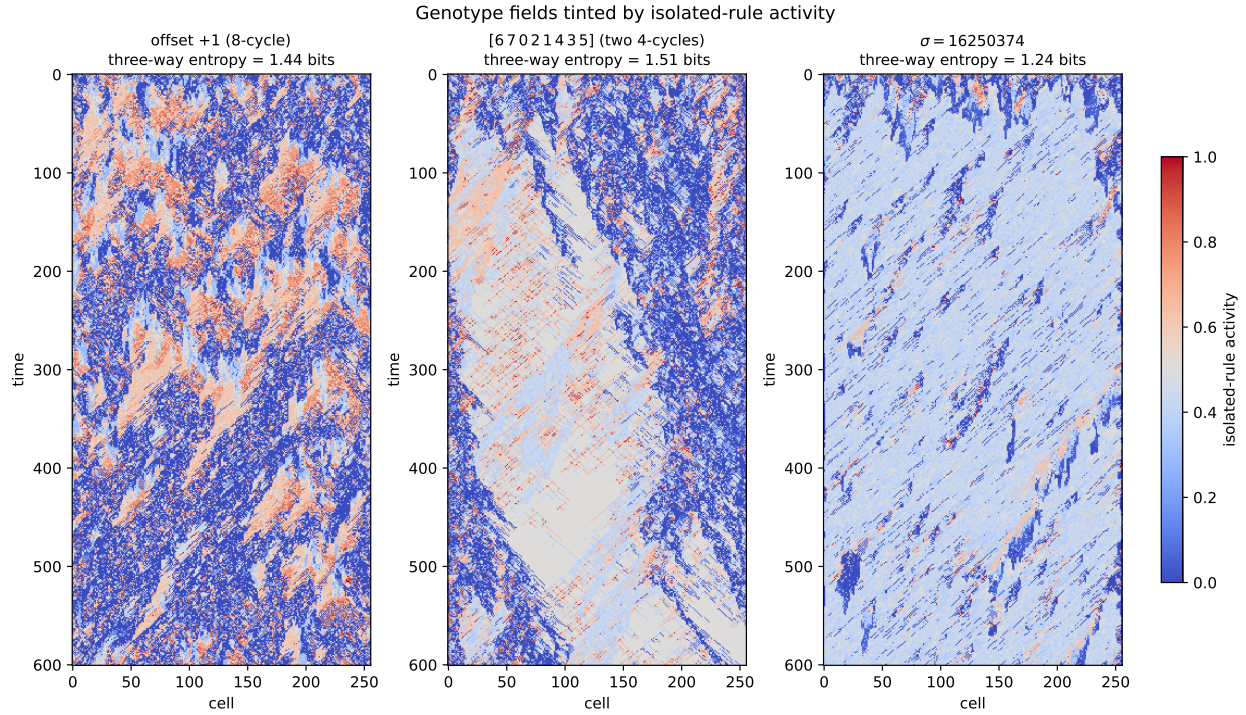
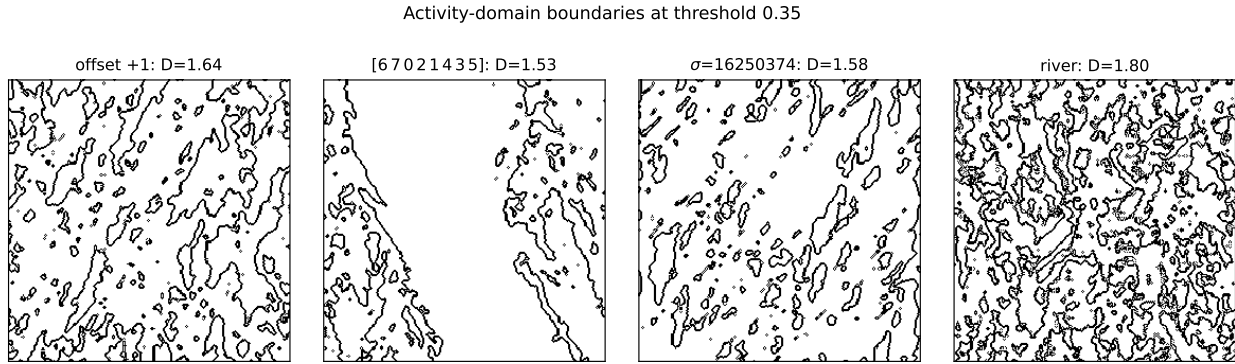


Figure 13

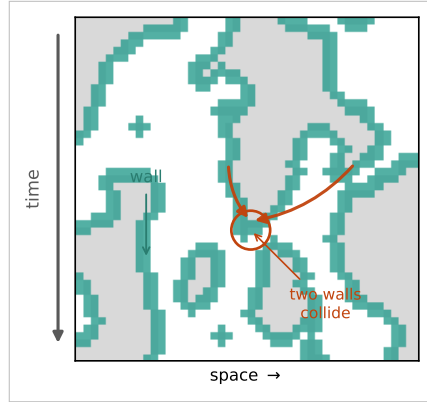
The genotype as a map of activity regimes. *Genotype spacetime fields*

(fixed-boundary variant) tinted by each rule's isolated-rule activity score (Definition 8; blue = low near 0, pale = intermediate near 0.4, red = high near 0.5 and above). Which active regime pairs with the low-activity background depends on the operator, and the three-way balance differs sharply (entropy in bits, in title): offset +1 (the 8-cycle survivor) mixes low with high-activity chaotic rules in a fine texture; the two-4-cycle [6 7 0 2 1 4 3 5]—the standard-order form of the historical offset+2 exemplar—forms large coherent intermediate-activity domains with sharp walls, the most balanced regime mix (entropy 1.51); and $\sigma = 16250374$ is dominated by the intermediate complex Rule 110. The score orders the regimes but does not by itself separate complex from chaotic.

**Figure 14**

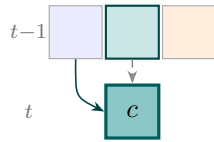
Activity-domain walls. The boundary (black) between low- and high-activity regions, traced through the 1+1-dimensional spacetime diagram at $L = 256$. Offset +1, the two-4-cycle [6 7 0 2 1 4 3 5], and $\sigma = 16250374$ carry fractal-like boundaries ($D \approx 1.5$ – 1.6); the phenotype-coupled river carries a denser one ($D \approx 1.80$, an unmatched comparison as it couples phenotype). The dimension is stable across $L = 128$ – 768 (offset +1 ≈ 1.67 , the two-4-cycle ≈ 1.50 , $\sigma \approx 1.54$, and river ≈ 1.79 ; box sizes 2–64). Collapsed operators (e.g. offset +4) carry none. These walls are also consistently more rewritten than the chaotic-domain interior (Discussion). Exploratory: a handful of seeds, fixed-boundary variant.

(a) a real patch (offset+1): activity domains and their walls



(b) transport

along a wall

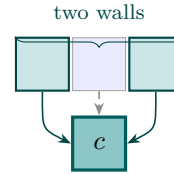
*transfer entropy:*

a neighbour's past

beyond c 's own

(c) combining

where walls collide

*synergy:*

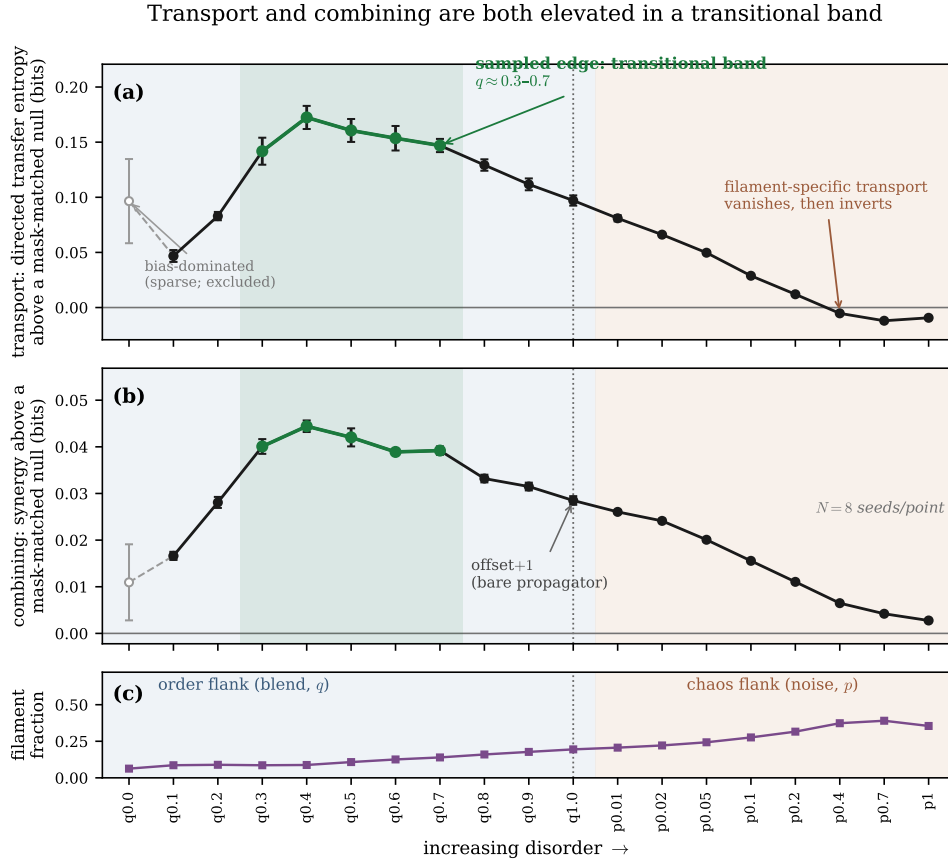
the two walls jointly,

beyond their

separate effects

Figure 15

How the two measures read a filament. (a) A real space–time patch from the synthetic family (offset+1): low- and high-activity domains (white/grey) meet along thin walls (teal), with one wall and one collision marked. In the schematics each cell c is predicted from its own past (dashed) and its two spatial neighbours' pasts (teal); tints mark the low- (blue) and high-activity (orange) domains a wall separates. (b) Transport is the directed transfer entropy from a neighbour into c beyond c 's own past, elevated on the walls. (c) Combining is the synergy of the two neighbours—their joint contribution beyond the sum of their separate ones—probed where two walls collide. History length one; conditioning on the own past removes what the wall merely carries forward in time.

**Figure 16**

A sampled edge of chaos. Both measures across a synthetic order–chaos family: the blend interrupter tunes toward order (fraction q), a seeded stochastic rule-flip tunes toward chaos (rate p), with the bare offset+1 propagator between them. **(a)** transport and **(b)** combining, each plotted as the filament-specific score—the on-filament value minus a mask-matched random draw of the same cell count—so that zero means the filament is no better than an arbitrary set of cells that size. Points are means over $N = 8$ seeds; bars are $\pm SEM$. Both are elevated across the shaded transitional band ($q \approx 0.3-0.7$) and decay monotonically into chaos; on the chaotic flank transport inverts while combining stays weakly positive. The open grey point at $q = 0$ is bias-dominated and is excluded from all claims. **(c)** the filament fraction, which is why the mask-matched null is needed: the extracted wall grows from 20% to 39% of the field. Test statistics, the unresolved band interior, and the single-family caveat are given in the text. Both nulls are seeded per field, so the reported values are run-invariant; reproducible from `mmca.experiments.eoc-sweep`.

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